

IDENTIFICATION

Product Code: Maindec-08-D71A-D
Product Name: 680 DCS Expanded Static Test
Date Created: May 15, 1967
Maintainer: Diagnostic Group
Author: E. Kenney

1 ABSTRACT

The 680 DCS Expanded Static Test consists of two independent test sequences intended to verify correct operation of the IOT instructions and control logic associated with the 680 Data Communications System. Test failures will result in a processor halt at a halt location associated with the failing test. Each test halt is interpreted as to failing test sequence, suggested failure source (where possible), and program restart instructions.

2 REQUIREMENTS

2.1 Storage

<u>Test</u>	<u>Starting Address</u>	<u>End Address</u>	
Line Clock Test	0200	0513	(Section 4)
IOT and Control Test	0674	2706	(Section 5)

2.2 Equipment

Minimum configuration PDP-8

Minimum configuration 680 DCS (i.e., 681 Data Line Interface and 685 Multiplexer Control)

3 PROGRAM LOADING

- If the Binary Loader is resident in memory proceed to step b. Otherwise load the Binary Loader into memory.
- Set the AC switch register to 7777 and depress the LOAD ADDRESS key. Then depress the START key.
- Place the 680 DCS Expanded Static Test in the keyboard reader and turn the reader on.
- When the binary program tape has been completely read into memory, the AC should contain zero indicating correct program tape checksum.

4 LINE CLOCK TEST

4.1 General Description

The line clock test verifies correct execution of the three IOT instructions associated with a specified clock number, and correct operation of the logic associated with each clock.

The three IOT Instructions are listed below:

<u>IOT</u>	<u>Octal Value</u>	<u>Operation</u>
TTXSKP	64Y1	Skip if specified clock flag is set
TTXON	64Y4	Reset and enable specified clock flag
TTXOFF	64Y2	Reset and disable specified clock flag

NOTE: X = 1, 2, 3, or 4 and Y = 2, 3, 4, or 5, respectively

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With SR 2 set, the time intervals in microseconds of eight successive clock interrupts will be typed on the Model 33 Teleprinter.

The following chart lists the correct interrupt time intervals for various clock speeds:

<u>Crystal Frequency</u>	<u>Line Baud Rate</u>	<u>Bit Time</u>	<u>Interrupt Time Interval (μsec)</u>
14.08 kc	110	9.09 ms	1136
9.6 kc	75	13.3 ms	1665
6.4 kc	50	20.0 ms	2500
5.76 kc	45	22.2 ms	2780

NOTE: The interrupt time intervals will be reported within a tolerance of 1 percent.

The following example is the interrupt time interval report for clock no. 1 (110 baud):

CLOCK 1
1132 1124 1132 1132 1124 1132 1132 1132

4.2 Operating Procedure

- a. Set SR to 0200 and depress LOAD ADDRESS
- b. Select number of clock to be tested (SR 9, 10, and 11):
Clock No. 1 - SR 11 set; SR 9 and 10 reset
Clock No. 2 - SR 10 set; SR 9 and 11 reset
Clock No. 3 - SR 10 and 11 set; SR 9 reset
Clock No. 4 - SR 9 set; SR 10 and 11 reset
- c. Select desired program control according to the following Switch Register Summary.

SR0 Set - Halt when error occurs
Reset - Bypass halt and continue to cycle

SR1 Set - Normal test sequence consists of
1. Reset and disable all clocks (TTXOFF)
2. Reset and enable specified clock (TTXON)
3. Enter Interrupt timer loop and wait for clock interrupt
4. Verify interrupt from specified clock and repeat steps 2 and 3, indefinitely (Interrupt time intervals are not stored or reported.)

Reset - Normal test sequence consists of
1. Reset and disable all clocks (TTXOFF)
2. Reset and enable specified clock (TTXON)
3. Enter interrupt time loop and wait for clock interrupt
4. Verify interrupt from specified clock and store interrupt time interval
Repeat steps 2, 3, and 4, eight times
5. After eight clock interrupts, reset and disable clock and return to interrupt timer loop to verify correct execution of the TTXOFF instruction
6. When the interrupt timer expires (4MS), the eight stored interrupt time intervals may be typed (SR 2 set) and the test sequence is restarted at step 1 (SR 3 reset)

- SR2 Set - Type eight successive clock interrupt time intervals
Reset - Bypass type routine
- SR3 Set - Halt at completion of clock test (step 6)
Reset - Continuously cycle through steps 1-6
- SR4 Set - Ring bell if error is detected
Reset - Bypass bell

d. Depress START. Clock test will now cycle. Note: Since the program stores the clock number specified in SR 9, 10, and 11 each time it recycles, the number of the clock to be tested may be changed while the clock test is cycling.

4.3 Description of Clock Test Error Halts

All clock test halts are referenced by an absolute octal memory address (halt location), and a mnemonic tag (halt tag). The Description of Halt associated with each program halt provides a statement of conditions leading to the halt and the program sequence following depression of CONTINUE.

NOTE: Except where otherwise specified, selection of bypass error halt (SR0 Reset) will result in the program sequence described for Continue.

<u>Halt Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
206	ECT1	Operator selected nonexistent clock number (i.e., neither clock no. 1, 2, 3, or 4 was specified). Select desired clock number and depress CONTINUE. (SR 0 will not cause this halt to be bypassed.)
072	ECT2	Interrupt resulting from incorrect clock flag. AC contains number of clock causing error. Depress CONTINUE to reset and disable all clock flags, enable the clock specified by SR 9, 10, and 11, and restart clock test. Note: If error halts are bypassed (SR 0 reset) the clock causing the error will be reset and disabled and the clock test will continue.
134	ECT3	Interrupt resulting from source other than the 680 clock flags (1, 2, 3, and 4) or the Model 33 Teleprinter flag. Depress CONTINUE to restart clock test at location 200.
257	ECT4	Specified clock did not generate an interrupt within 4 ms after being enabled. Depress CONTINUE to restart clock test at location 200.

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<u>Halt Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
466	ECT5	Specified clock generated an interrupt within 4 ms after being disabled. Depress CONTINUE to reset and disable clock and reenter interrupt timer.
512	ECT	End of clock test (SR 3 set). Depress CONTINUE to restart clock test at location 200.

5 IOT AND CONTROL TEST

5.1 General Description

The IOT and control test is divided into 3 groups of related subtests involving:

- a. 681 and 685 device selection (40 and 41, respectively).
- b. IOT instructions (listed below) line selection register (LSR) and control logic associated with the 685.

<u>IOT</u>	<u>Octal Value</u>	<u>Operation</u>
TTCL	6411	Clear the LSR to 0
TTSL	6412	Load LSR from AC and clear AC to 0 (AC 5-11 inclusive or to LSR 0-6)
TTRL	6414	Load AC from LSR (LSR 0-6 inclusive OR to AC 5-11)
TTINCR	6401*	Increment the LSR by 1

*Decoded in 681

- c. IOT instructions (listed below) and control logic associated with the 681 Data Line Interface.

<u>IOT</u>	<u>Octal Value</u>	<u>Operation</u>
TTI	6402	Transfer data from the 681 to the computer's MB register
TTO	6404	Transfer data from the AC to the 685

Execution of the IOT and control test does not require any connection of input data lines to output data lines. (Line no. 177 will be addressed during any tests requiring a mark level on the input data line.)

5.2 Operating Procedure

- a. Set SR to 0674 and depress LOAD ADDRESS.
- b. Select desired program control according to the following Switch Register Summary:

SR0	Set - Halt when error occurs
	Reset - Disregard error, bypass halt and continue to cycle
SR1	Set - Scope mode (don't test for error, loop in current test)
	Reset - Normal test cycle

- SR2 Set - Verify mode (test for errors, cycle in current test)
Reset - Cycle through all tests
- SR3 Set - Halt at completion of all tests
Reset - Repeat all tests
- SR4 Set - Ring bell if error is detected
Reset - Bypass bell
- SR5 Set - Ring bell after 680 complete test cycles (approximately 10 sec.)
Reset - Bypass bell

c. Depress START - The IOT and control test will be executed

5.3 Description of IOT and Control Test Error Halts

All test halts are referenced by an absolute octal memory address (halt location) and a mnemonic tag (halt tag). The Description of Halt associated with each program halt provides a statement of conditions leading to the halt and the program sequence following depression of CONTINUE.

NOTE: All 680 clocks are disabled during execution of the IOT and control test; the interrupt system is enabled (ION).

a. All error halts are listed below their associated subtest mnemonic and starting address.

NOTE: DST = 681-685 Device selection test group
LSR = 685 Line selection register test group
TS681 = 681 IOT and control test group

<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
160	BADINT	Interrupt occurred from source other than the Model 33 Teleprinter flag. Depress CONTINUE to clear and disable all 680 clock flags and return to the particular test in progress. (Note: The return address is stored in location 0)
		Test DSTA - (location 674)
722	ERDST1	AC before execution = 4000; the 685 is addressed via IOT 6414 (TTRL). Expected contents of AC = 4000; actual results are displayed in the AC IOT 6414 should affect AC 5-11 only. If resulting AC = 2000, IOT 6414 may have been decoded in 681 as IOT 6404 Depress CONTINUE to repeat test DSTA

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<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
		Test DSTB - (location 724)
746	ERDST2	AC before execution = 0000; the 681 is addressed via IOT 6404 (TTO). Expected contents of AC = 0000; actual results are displayed in the AC If AC 5-11 $\neq$ 0, IOT 6404 may have been decoded in 685 as IOT 6414 Depress CONTINUE to repeat test DSTB Test LSRA - (location 1000)
1017	ERLSR1	AC before execution = 7777; the 685 is addressed via IOT 6412 (TTSL). Expected contents of AC = 0000; actual results are displayed in the AC IOT 6412 should clear the AC to 0 at IOP2 time Depress CONTINUE to repeat test LSRA Test LSRB - (location 1023)
1044	ERLSR2	AC before execution = 7777; the 685 is addressed via IOT 6411 (TTCL). Expected contents of AC = 7777; actual results are displayed in the AC IOT 6411 should not effect the AC Depress CONTINUE to repeat test LSRB Test LSRC - (location 1046)
1067	ERLSR3	AC before execution = 7777; the 685 is addressed via IOT 6414 (TTRL). Expected contents of AC = 7777; actual results are displayed in the AC The inclusive OR transfer of the LSR to AC should not affect the contents of the AC (regardless of the contents of the LSR) Depress CONTINUE to repeat test LSRC Test LSRD - (location 1071)
1113	ERAC1	IOT 6412 (TTSL) associated with this subtest failed to clear the AC Depress CONTINUE to reexecute test LSRA
1121	ERLSR4	With the AC preset to 7777, IOT 6412 (TTSL) is issued to load the LSR with 177 and clear the AC to 0 With the AC = 0000, IOT 6414 (TTRL) is issued to load the AC with the contents of the LSR. Expected contents of AC = 0177; actual results are displayed in the AC. Error could be a result of incorrect execution of IOT 6412 or IOT 6414, or both Depress CONTINUE to repeat test LSRD

<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
		Test LSRE - (location 1125)
1156	ERAC2	IOT 6412 (TTSL) associated with this subtest failed to clear the AC Depress CONTINUE to reexecute test LSRA
1164	ERLSR5	The LSR is loaded with 177 twice, via two sequential IOT 6412 instructions (TTSL). With the AC = 0000, IOT 6414 (TTRL) is issued to load the AC with the contents of LSR. Expected contents of LSR and AC = 0177; actual results are displayed in the AC. Error indicates problem in inclusive OR gating of LSR flip-flops Depress CONTINUE to repeat test LSRE
		Test LSRF - (location 1200)
1227	ERAC3	IOT 6412 (TTSL) associated with this subtest failed to clear the AC Depress CONTINUE to reexecute test LSRA
1235	ERLSR6	The LSR is sequentially loaded with 177 and 000 via two IOT 6412 (TTSL) instructions With the AC = 0000, IOT 6414 (TTRL) is issued to load the AC with the contents of the LSR. Expected contents of LSR and AC = 0177; actual results are displayed in the AC Error indicates problem in inclusive OR gating of LSR flip-flops Depress CONTINUE to repeat test LSRF
		Test LSRG - (location 1240)
1262	ERAC4	IOT 6412 (TTSL) associated with this subtest failed to clear the AC Depress CONTINUE to reexecute test LSRA
1267	ERLSR7	The LSR is loaded with 177 via IOT 6412 (TTSL). IOT 6411 (TTCL) and 6414 (TTRL) are sequentially issued to reset the LSR to 0 and transfer the LSR to the AC Expected contents of LSR and AC = 000; actual results are displayed in the AC Error indicates incorrect execution of IOT 6411 (TTCL) Depress CONTINUE to repeat test LSRG
		Test LSRH - (location 1271)
1315	ERAC5	IOT 6412 (TTSL) associated with this subtest failed to clear the AC Depress CONTINUE to reexecute test LSRA

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<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
1323	ERLSR8	The LSR is cleared to 0 via IOT 6411 (TTCL). With the AC = 0125, IOT 6412 (TTSL) is issued to set LSR 0, 2, 4, and 6 to 1. IOT 6414 (TTRL) is then issued to load the AC with the contents of the LSR Expected contents of LSR and AC = 125; actual results are displayed in the AC Depress CONTINUE to repeat test LSRH
1353	ERAC6	Test LSRI - (location 1327) IOT 6412 (TTSL) associated with this subtest failed to clear the AC Depress CONTINUE to reexecute test LSRA
1361	ERLSR9	The LSR is cleared to 0 via IOT 6411 (TTCL). With the AC = 0052, IOT 6412 (TTSL) is issued to set LSR 1, 3 and 5 to 1. IOT 6414 (TTRL) is then issued to load the AC with the contents of the LSR Expected contents of LSR and AC = 052; actual results are displayed in the AC Depress CONTINUE to repeat test LSRI Test LSRJ - (location 1400)
1431	ELSR10	The LSR is sequentially cleared, loaded and read for all bit configurations (000-177) via IOT 6417 The expected contents of LSR and AC following execution of IOT 6417 are displayed in the AC Depress CONTINUE to cause the actual results to be displayed in the AC Depress CONTINUE again to repeat test LSRJ with the failing bit configuration Note: Selection of scope mode (SR1) causes the error check to be bypassed and the current bit configuration to be cycled continuously Selection of verify mode (SR2) allows the error check to be performed and all bit configurations (000-177) to be cycled continuously Test LSRK - (location 1441)
1500	ELSR11	The LSR is initially cleared to 0 via IOT 6411 (TTCL) The LSR is now incremented from 001 through 000, via successive IOT 6401 (TTINCR) instructions. The contents of the LSR are read and compared following each IOT 6401 instruction The expected contents of the LSR are displayed in the AC Depress CONTINUE to cause the actual results to be displayed in the AC Depress CONTINUE again to repeat test LSRK for the failing bit configuration

<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
Note: Selection of scope mode (SR1) causes the error check to be bypassed and the current bit configuration to be cycled continuously Selection of verify mode (SR2) allows the error check to be performed and the LSR to be continuously incremented from 001 through 000 TS681A - (location 1600) The 681 Data line Interface is addressed via IOT 6400 (NOP). The 681 should take no action and the next instruction in sequence should be executed.		
1625	ERTTO1	The instruction immediately following IOT 6400 was not executed Error could be a result of incorrect generation of skip enable Depress CONTINUE to repeat test TS681A
1631	ERTTO2	Two instructions immediately following IOT 6400 were not executed Error indicates that IOT 6400 was incorrectly interpreted as IOT 6402 (TTI) Depress CONTINUE to repeat test TS681A
1635	ERTTO3	The link bit was cleared to 0 indicating that IOT 6400 was incorrectly interpreted as IOT 6404 (TTO) Depress CONTINUE to repeat test TS681A TS681B - (location 1641) With the AC = 0000 and the Link = 1, IOT 6404 (TTO) is issued. Following execution of IOT 6404 the AC and Link should contain 0.
1663	ERTTO4	The Link contained 1 following execution of IOT 6404 The resulting contents of the AC and Link are displayed Check operation of zero Link and RAR in the 681 Depress CONTINUE to repeat test TS681B
1670	ERTTO5	The Link was reset but AC0 was set to 1. The resulting contents of the Link and AC are displayed Check operation of zero Link Depress CONTINUE to repeat test TS681B TS681C - (location 1672)
1715	ERTTO6	With the AC = 4001 and Link = 1, IOT 6404 (TTO) is issued. If IOT 6404 is executed correctly the Link will be cleared to 0, AC 11 will be inhibited from shifting into the Link and AC 0-10 will be shifted one place to the right. The expected test results are AC = 2000 and Link = 0; the actual test results are displayed

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<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>										
1723	ERTTO7	Link was not reset to 0. Assuming TS681B was executed correctly, this error indicates failure of Link rotate disable Depress CONTINUE to repeat test TS681C AC did not contain 2000 indicating incorrect execution of RAR Depress CONTINUE to repeat test TS681C TS681D - (location 1727) With the AC = 5252 and Link = 0, IOT 6404 (TTO) is issued. Expected test results are AC = 2525 and Link = 0. The actual test results are displayed										
1751	ERTTO8	Link was set to 1. Check for incorrect execution of CML Depress CONTINUE to repeat test TS681D										
1760	ERTTO9	AC did not contain 2525 following execution of IOT 6404. Check execution of RAR Depress CONTINUE to repeat test TS681D TS681E - (location 2000) This test verifies correct execution format of IOT 6402 (TTI) IOT 6402 instruction format: <table><tr><td>Location A (6402)</td><td>TTI Instruction</td></tr><tr><td>A+1 (LSW)</td><td>Line Status Word</td></tr><tr><td>A+2 (CAW)</td><td>Character Assembly Word</td></tr><tr><td>A+3 (XXXX)</td><td>Next Instruction in Sequence</td></tr><tr><td>A+4 (XXXX)</td><td></td></tr></table>	Location A (6402)	TTI Instruction	A+1 (LSW)	Line Status Word	A+2 (CAW)	Character Assembly Word	A+3 (XXXX)	Next Instruction in Sequence	A+4 (XXXX)	
Location A (6402)	TTI Instruction											
A+1 (LSW)	Line Status Word											
A+2 (CAW)	Character Assembly Word											
A+3 (XXXX)	Next Instruction in Sequence											
A+4 (XXXX)												
2020	ERTTI1	The contents of A+1 were incorrectly interpreted as an instruction. Check for generation of the S cycle and its associated control functions (e.g., TTSET, PC → MA, Enable, Spec. cycle) Depress CONTINUE to repeat test TS681E										
2024	ERTTI2	Location A+1 was correctly interpreted as the LSW but A+2 was incorrectly interpreted as an instruction. Check for correct operation of skip bus in enable during the S cycle Depress CONTINUE to repeat test TS681E										
2030	ERTTI3	The instruction in location A+4 was the first instruction executed after completion of the TTI instruction. Since A+3 was not executed check for incorrect (extra) skip enable during the S cycle Depress CONTINUE to repeat test TS681E										

<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
		TS681F - (location 2034) See IOT 6402 format (pg. 10)
		With the LSW and CAW preset to 2000, IOT 6402 (TTI) is issued
		Correct execution of IOT 6402 should result in the contents of LSW shifted one position to the right with LSW0 = 0 (resulting LSW = 1000). The CAW should be unmodified (resulting CAW = 2000)
		Note: The LSR is loaded with 177 before the test is executed
2073	ERTTI4	Resulting LSW $\neq$ 1000. Actual results are displayed in the AC. Check operation of shift MB Depress CONTINUE to repeat test TS681F
2104	ERTTI5	The LSW was shifted correctly but LSW0 = 1 indicates a start level on the Teletype data in lead. Check operation of MB0 shift enable. (Resulting LSW is displayed in the AC) Depress CONTINUE to repeat test TS681F
2115	ERTTI6	LSW 9, 10, and 11 were $\neq$ 0 following execution of IOT 6402 indicating incorrect execution of count MB Depress CONTINUE to repeat test TS681F
2121	ERTTI7	Resulting LSW $\neq$ 1000. Actual results are displayed in the AC. Check for a combination of the above three error conditions Depress CONTINUE to repeat test TS681F
2127	ERTTI8	CAW was modified; results are displayed in the AC Check for incorrect generation of C cycle. (C cycle should be executed only when LSW0 = 1 and LSW 9, 10, and 11 = 0, 1, 1, respectively) Depress CONTINUE to repeat test TS681F
		TS681G - (location 2200) See IOT 6402 format (pg. 10)
		With the CAW preset to 4000 and the LSW present to the condition required for entry into the C cycle (LSW = 4003), IOT 6402 (TTI) is issued. Correct execution of IOT 6402 should result in LSW = 4004 and CAW = 6000
		Note: The LSR is preloaded with 177
2234	ERTTI9	LSW $\neq$ 4004. Check operation of count MB, shift MB and MB0 shift enable. (The actual LSW results are displayed in the AC) Depress CONTINUE to repeat test TS681G

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<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
2246	ETTI10	CAW $\neq$ 6000. (Actual CAW results are displayed in the AC). Check C cycle execution (e.g., shift MB, MB0 shift enable, TTSET, etc.) Depress CONTINUE to repeat test TS681G
2252	ETTI11	CAW $\neq$ 6000, contents of location A+3 were modified (Actual results of CAW are displayed in the AC) Check for incorrect skip enable during the S cycle Depress CONTINUE to repeat test TS681G TS681H - (location 2262) See IOT 6402 format, (pg. 10) IOT 6402 (TTI) is issued with successive LSW counts of 4, 5, 6, 7, 0, 1 and 2 (i.e., 4004 - 4002) The LSW and CAW are examined after each TTI execution to insure correct execution of count MB and correct LSW count interpretation Note: The LSR is preloaded with 177
2340	ETTI12	LSW0 was reset to 0 (resulting LSW is displayed in AC) Check operation of S cycle MB shift Depress CONTINUE to repeat test TS681H
2347	ETTI13	The LSW count was not correct following execution of IOT 6402. The expected contents of the LSW are displayed in the AC Depress CONTINUE to display the actual LSW results Depress CONTINUE again to retest failing count
2361	ETTI14	The CAW was modified indicating incorrect entry into C cycle. (Expected contents of CAW = 4000, actual CAW results displayed in AC) Depress CONTINUE to retest failing count Note: Selection of scope mode (SR1) causes the error check to be bypassed and the current LSW count to be cycled continuously Selection of verify mode (SR2) allows error check to be performed and LSW counts 4 through 2 to be continuously tested TS681I - (location 2400) This test verifies correct operation of MB0 (1) shift enable and MB1-11 (1) shift
2440	ERMBS1	Incorrect MB shift results. Expected results are displayed in the AC Depress CONTINUE to display the actual MB shift results in the AC Depress CONTINUE again to retest failing MB shift Note: Selection of scope mode (SR1) causes the MB shift error check to be bypassed and the current MB shift configuration to be cycled continuously

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<u>Octal Location</u>	<u>Halt Tag</u>	<u>Description of Halt</u>
		Selection of verify mode (SR2) allows the MB shift error check to be performed and the full MB shift test to be cycled continuously TS681J - (location 2456) This test verifies correct operation of MB1-11 (0) shift
2517	ERMBS2	Incorrect MB shift results. Expected results are displayed in the AC Depress CONTINUE to display the actual MB shift results in the AC Depress CONTINUE again to retest failing MB shift (See note for TS681I) TS681K - (location 2600) This test verifies correct operation of MB shift via a shift pattern of alternate 1s and 0s
2654	ERMBS3	Incorrect MB shift results. Expected results are displayed in the AC Depress CONTINUE to display the actual MB shift results in the AC Depress CONTINUE again to retest failing MB shift (See note for TS681I)
2674	ENDTST	End of Static Test. Depress CONTINUE to repeat all tests.

6 MISCELLANEOUS

It is suggested that the IOT and Control Test be run with appropriate voltage margins before attempting to execute the 680 DCS Data and Control Test (MAINDEC-08-D72A-D).

7 PROGRAM LISTING

/680 DCS EXPANDED STATIC TEST

/IOT INSTRUCTION DEFINITIONS

TT10FF=6422
TT20FF=6432
TT30FF=6442
TT40FF=6452

TT10N=6424
TT20N=6434
TT30N=6444
TT40N=6454

TT1SKP=6421
TT2SKP=6431
TT3SKP=6441
TT4SKP=6451

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/SWITCH REGISTER SETTINGS FOR IOT AND CONTROL TEST:

/

/SR0 - HALT ON ERROR
 /SR1 - SCOPE MODE (DON'T TEST FOR ERROR, LOOP IN CURRENT TEST)
 /SR2 - VERIFY MODE (TEST FOR ERRORS, LOOP IN CURRENT TEST)
 /SR3 - HALT AT COMPLETION OF STATIC TEST
 /SR4 - SIGNAL ERROR WITH BELL
 /SR5 - SIGNAL COMPLETION OF 4096 TEST CYCLES:

/

/SWITCH REGISTER SETTINGS FOR 680 CLOCK TEST:

/SR0 - HALT ON ERROR
 /SR1 - SCOPE MODE
 /SR2 - TYPE 8 SUCCESSIVE INTERRUPT TIME INTERVALS, (SR1 RESET)
 /SR3 - HALT AT COMPLETION OF CLOCK TEST
 /SR4 - SIGNAL ERROR WITH BELL

/PAGE ZERO: CONSTANTS AND SUBROUTINES

*0

0000	0000	0		
0001	3010	INTRPT,	DCA KEEPAC	/STORE AC
0002	7010	PAR		
0003	3011	DCA STORL		/STORE LINC
0004	5142	JMP CHTPTR		/TEST FOR TELEPRINTER FLAG
0005	0345	CLSRR,	JMPCL1-1	
0006	0000	COUNT,	0	/CONTAINS CLOCK INTERRUPT COUNT IN MSEC
0007	0000	FSTPAS,	0	/EQUALS ZERO TO SIGNIFY 1ST CLOCK INTERRUPT
0010	0000	KEEPAC,	0	
0011	0000	STORL,	0	
0012	0000	INTSTA,	0	
0013	0200	MSKSR4,	0200	
0014	0207	BELL,	0207	
0015	4000	MSKSR0,	4000	

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*0020
/DECODE SWITCH REGISTER 0
0020 0000 SR0, 0
0021 7604 CLA MSR
0022 0015 AND MSKSR0
0023 7640 SZA CLA
0024 2020 ISZ SR0 /ADD ONE TO RETURN ADDRESS IF SR0 IS SET
0025 7604 CLA MSR
0026 0013 AND MSKSR4
0027 7640 SZA CLA /RING BELL?
0030 5032 JMP .+2 /YES
0031 5420 JMP I SR0 /NO
0032 6002 IOF
0033 1014 TAD BELL
0034 6046 TLS
0035 6041 6041 /SKIP ON TELEPRINTER FLAG
0036 5035 JMP .-1
0037 7300 CLA CLL
0040 6001 ION
0041 5420 JMP I SR0

/DECODE SWITCH REGISTER 1
0042 0000 SR1, 0
0043 7604 CLA MSR
0044 0050 AND MSKSR1
0045 7640 SZA CLA
0046 2042 ISZ SR1 /ADD ONE TO RETURN ADDRESS IF SR1 IS SET
0047 5442 JMP I SR1
0050 2000 MSKSR1, 2000

/DECODE SWITCH REGISTER 2
0051 0000 SR2, 0
0052 7604 CLA MSR
0053 0057 AND MSKSR2
0054 7640 SZA CLA
0055 2051 ISZ SR2 /ADD ONE TO RETURN ADDRESS IF SR2 IS SET
0056 5451 JMP I SR2
0057 1000 MSKSR2, 1000

/DECODE SWITCH REGISTER 3
0060 0000 SR3, 0
0061 7604 CLA MSR
0062 0066 AND MSKSR3
0063 7640 SZA CLA
0064 2060 ISZ SR3 /ADD ONE TO RETURN ADDRESS IF SR3 IS SET
0065 5460 JMP I SR3
0066 0400 MSKSR3, 0400

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/COME HERE IF WRONG CLOCK INTERRUPTS

```

0067 4020 ERCL1,      JMS SR0          /DO WE WISH TO HALT? (SR0 SET)
0070 5074      JMP .+4                /DON'T HALT
0071 1126      TAD ONE
0072 7402 ECT2,      HLT              /HALT AND DISPLAY ERROR CLOCK NUMBER
0073 5477      JMP I ARSTRT          /RE-INITIALIZE CLOCK TEST
0074 6422      TT10FF                /RESET AND DISABLE CLOCK FLAG1
0075 5500      JMP I AINTON          /RETURN TO TIMER
0076 0232 ACLFLG,    CLRFLG
0077 0200 ARSTRT,    SELECT
0100 0233 AINTON,    INTON-1

0101 4020 ERCL2,      JMS SR0          /DO WE WISH TO HALT? (SR0 SET)
0102 5106      JMP .+4                /DON'T HALT
0103 1127      TAD TWO
0104 5072      JMP ECT2              /HALT AND DISPLAY ERROR CLOCK NUMBER
0105 5477      JMP I ARSTRT          /RE-INITIALIZE CLOCK TEST
0106 6432      TT20FF                /RESET AND DISABLE CLOCK FLAG2
0107 5500      JMP I AINTON          /RETURN TO TIMER

0110 4020 ERCL3,      JMS SR0          /DO WE WISH TO HALT? (SR0 SET)
0111 5115      JMP .+4                /DON'T HALT
0112 1130      TAD THREE
0113 5072      JMP ECT2              /HALT AND DISPLAY ERROR CLOCK NUMBER
0114 5477      JMP I ARSTRT          /RE-INITIALIZE CLOCK TEST
0115 6442      TT30FF                /RESET AND DISABLE CLOCK FLAG 3
0116 5500      JMP I AINTON          /RETURN TO TIMER

0117 4020 ERCL4,      JMS SR0          /DO WE WISH TO HALT? (SR0 SET)
0120 5124      JMP .+4                /DON'T HALT
0121 1131      TAD FOUR
0122 5072      JMP ECT2              /HALT AND DISPLAY ERROR CLOCK NUMBER
0123 5477      JMP I ARSTRT          /RE-INITIALIZE CLOCK TEST
0124 6452      TT40FF                /RESET AND DISABLE CLOCK FLAG 4
0125 5500      JMP I AINTON          /RETURN TO TIMER

0126 0001 ONE,       0001
0127 0002 TWO,       0002
0130 0003 THREE,    0003
0131 0004 FOUR,     0004

```

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```

/INTERRUPT FROM UNDETERMINED SOURCE
0132 4020 NOFLAG,      JMS SR0      /DO WE WISH TO HALT?
0133 5477     JMP I ARSTRT      /DON'T HALT
0134 7402 ECT3,        HLT        /UNDETERMINED INTERRUPT SOURCE
0135 5477     JMP I ARSTRT

0136 0000 TIMEOUT,     0          /CLOCK INTERRUPT AND TIME-OUT STATUS
0137 0000 CLOCK,       0          /CONTAINS NUMBER OF CLOCK BEING TESTED
0140 0000 STORAC,      0
0141 0000 AUXAC,       0

0142 6041 CHTPTR,      TSF        /SKIP IF TELEPRINTER FLAG SET
0143 5153     JMP CHKINT
0144 6042     TCF              /CLEAR TELEPRINTER FLAG

/RESUME NORMAL PROGRAM SEQUENCE
0145 7300 RNPS,        CLA CLL
0146 1011     TAD STORL
0147 7004     RAL              /RESTORE LINC BIT
0150 1010     TAD KEEPAC      /RESTORE AC
0151 6001     ION
0152 5400     JMP I INTRPT-1

/TEST INTERRUPT STATUS
0153 1136 CHKINT,      TAD TIMEOUT
0154 7500     SMA              /SKIP IF INTERRUPT ERROR HAS OCCURRED
0155 5405     JMP I CLSRR      /TEST FOR SPECIFIED CLOCK FLAG
0156 4020     JMS SR0         /HALT ON ERROR?
0157 5161     JMP .+2         /NO
0160 7402 RADINT,      HLT      /HALT DUE TO INTERRUPT ERROR

/CLEAR AND DISABLE ALL 680 CLOCK FLAGS
0161 6422     TT1OFF
0162 6432     TT2OFF
0163 6442     TT3OFF
0164 6452     TT4OFF
0165 5145     JMP RNPS

```

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/680 DCS EXPANDED STATIC TEST

/LINE CLOCK TEST

*200

/OPERATOR MUST SPECIFY CLOCK NUMBER 1, 2, 3 OR 4

```

0200 7604 SELECT,      CLA OSR                /BRING S.R. INTO A.C.
0201 7342     AND MASK1                /DISREGARD AC0-AC8
0202 3137     DCA Z CLOCK              /STORE CLOCK SELECTION
0203 1137     TAD Z CLOCK
0204 7440     SZA                      /VERIFY THAT CLOCK 1-4 IS SELECTED
0205 5210     JMP .+3
0206 7402     ECT1,      HLT            /OPERATOR SELECTED NON EXISTANT CLOCK
0207 5200     JMP SELECT
0210 1343     TAD M5
0211 7500     SMA                      /SKIP ON MINUS AC
0212 5206     JMP .-4

```

/OPERATOR SELECTED EXISTANT CLOCK-PROCEED WITH TEST

/SET UP CLOCK SUBROUTINE ADDRESS POINTER

```

0213 7300     RESTRT,     CLA CLL
0214 1345     TAD JMPADD
0215 1137     TAD Z CLOCK
0216 3005     DCA Z CLSUBR            /CLSUBR NOW POINTS TO CORRECT CLOCK
                                         /SUBROUTINE
0217 3136     DCA Z TIMEOUT          /RESET TIME-OUT ADDRESS POINTER
0220 3007     DCA FSTPAS             /RESET FIRST PASS INDICATOR

```

/RESET AND DISABLE ALL CLOCK FLAGS, (64X2)

```

0221 6422     TT10FF
0222 6432     TT20FF
0223 6442     TT30FF
0224 6452     TT40FF

```

/ENABLE SELECTED CLOCK

```

0225 1137     TAD Z CLOCK
0226 7106     CLL RTL
0227 7004     RAL
0230 1352     TAD ALLCLK
0231 3232     DCA CLRFLG
0232 6424     CLRFLG,      6424        /THIS INSTRUCTION IS MODIFIED TO ENABLE
0233 1007     TAD Z FSTPAS            /SPECIFIED CLOCK
0234 6001     INTON,      ION          /ENABLE THE INTERRUPT SYSTEM
0235 7640     SZA CLA
0236 5241     JMP TIMER+1            /DON'T RESET COUNT IF FSTPAS IS ONE
0237 5240     JMP TIMER

```

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/CLOCK INTERRUPT TIMER

```

0240 3006  TIMER,          DCA COUNT                      /RESET INTERRUPT TIMER
0241 1354      TAD M412
0242 3353      DCA PASS                      /PRESET INTERRUPT TIME OUT COUNTER
0243 1006  ADD,          TAD COUNT
0244 1355      TAD TIME                      /TIME EQUALS 15 MICROSECONDS (DECIMAL)
0245 3006      DCA COUNT
0246 7000      NOP
0247 2353      ISZ PASS                      /GO TO TIME OUT ROUTINE IF PASS EQUALS 0
0250 5243      JMP ADD

```

/COME HERE IF INTERRUPT TIME OUT RESULTS

```

0251 6002      IOF                      /DISABLE INTERRUPT SYSTEM
0252 1136      TAD Z TIMEOUT              /EXAMINE TIME OUT STATUS INDICATOR
0253 7440      SZA
0254 5744      JMP I AEXIT              /TIME OUT EXPECTED

```

/TIMEOUT NOT EXPECTED

```

0255 4020      JMS Z SR0                      /DO WE HALT ON ERRORS? (S.R.0 SET)
0256 5213      JMP RESTRT                  /NO! RESTART TEST
0257 7402      ECT4,          HLT          /NO INTERRUPT FROM SELECTED CLOCK
                                /OPERATOR MAY RESTART TEST BY DEPRESSING CONTINUE
0260 5213      JMP RESTRT

```

/INTERRUPT SUBROUTINE-CLOCK1

```

0261 6431  CLOCK1,      TT2SKP                      /SKIP IF WRONG CLOCK FLAG SET (CLOCK 1)
0262 5264      JMP .+2
0263 5101      JMP Z ERCL2                      /CLOCK 2 FLAG SET
0264 6441      TT3SKP                      /SKIP IF WRONG CLOCK FLAG SET, (CLOCK 3)
0265 5267      JMP .+2
0266 5110      JMP Z ERCL3                      /CLOCK 3 FLAG SET
0267 6451      TT4SKP                      /SKIP IF WRONG CLOCK FLAG SET, (CLOCK 4)
0270 5272      JMP .+2
0271 5117      JMP Z ERCL4                      /CLOCK 4 FLAG SET
                                /VERIFY THAT SPECIFIED CLOCK CAUSED INTERRUPT
0272 6421      TT1SKP
0273 5132      JMP Z NOFLAG                  /NO CLOCK FLAG SET
0274 5741      JMP I CLOKOK                  /INTERRUPT CAUSED BY CORRECT CLOCK

```

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```

/INTERRUPT SUBROUTINE - CLOCK 2
0275 6421 CLOCK2, TT1SKP /SKIP IF WRONG CLOCK FLAG SET, (CLOCK1)
0276 5300 JMP .+2
0277 5067 JMP Z ERCL1 /CLOCK 1 FLAG SET
0300 6441 TT3SKP /SKIP IF WRONG CLOCK FLAG SET (CLOCK 3)
0301 5303 JMP .+2
0302 5110 JMP Z ERCL3 /CLOCK 3 FLAG SET
0303 6451 TT4SKP /SKIP IF WRONG CLOCK FLAG SET (CLOCK 4)
0304 5306 JMP .+2
0305 5117 JMP Z ERCL4 /CLOCK 4 FLAG SET
/VERIFY THAT SPECIFIED CLOCK CAUSED INTERRUPT
0306 6431 TT2SKP
0307 5132 JMP Z NOFLAG /NO CLOCK FLAG SET
0310 5741 JMP I CLOKOK /INTERRUPT CAUSED BY CORRECT CLOCK

/INTERRUPT SUBROUTINE - CLOCK 3
0311 6421 CLOCK3, TT1SKP /SKIP IF WRONG CLOCK FLAG SET, (CLOCK 1)
0312 5314 JMP .+2
0313 5067 JMP Z ERCL1 /CLOCK 1 FLAG SET
0314 6431 TT2SKP /SKIP IF WRONG CLOCK FLAG SET, (CLOCK 2)
0315 5317 JMP .+2
0316 5101 JMP Z ERCL2 /CLOCK 2 FLAG SET
0317 6451 TT4SKP /SKIP OF WRONG CLOCK FLAG SET (CLOCK 4)
0320 5322 JMP .+2
0321 5117 JMP Z ERCL4 /CLOCK 4 FLAG SET
/VERIFY THAT SPECIFIED CLOCK CAUSED INTERRUPT
0322 6441 TT3SKP
0323 5132 JMP Z NOFLAG /NO CLOCK FLAG SET
0324 5741 JMP I CLOKOK /INTERRUPT CAUSED BY CORRECT CLOCK

/INTERRPT SUBROUTINE - CLOCK 4
0325 6421 CLOCK4, TT1SKP /SKIP IF WRONG CLOCK FLAG SET (CLOCK )
0326 5330 JMP .+2
0327 5067 JMP Z ERCL1 /CLOCK 1 FLAG SET
0330 6431 TT2SKP /SKIP IF WRONG CLOCK FLAG SET (CLOCK 2)
0331 5333 JMP .+2
0332 5101 JMP Z ERCL2 /CLOCK 2 FLAG SET
0333 6441 TT3SKP /SKIP IF WRONG CLOCK FLAG SET (CLOCK 3)
0334 5336 JMP .+2
0335 5110 JMP Z ERCL3 /CLOCK 3 FLAG SET
/VERIFY THAT SPECIFIED CLOCK CAUSED INTERRUPT
0336 6451 TT4SKP
0337 5132 JMP Z NOFLAG /NO CLOCK FLAG SET
0340 5741 JMP I CLOKOK /INTERRUPT CAUSED BY CORRECT CLOCK

0341 6400 CLOKOK, COMCLK

```

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```

0342 0007 MASK1,      0007
0343 7773 M5,        -5
0344 0470 AEXIT,     EXIT

      /INTERRUPT ADDRESS POINTER
0345 0345 JMPADD,     JMPCL1-1

      /JUMP TO SPECIFIED CLOCK SUBROUTINE
0346 5261 JMPCL1,     JMP CLOCK1
0347 5275     JMP CLOCK2
0350 5311     JMP CLOCK3
0351 5325     JMP CLOCK4

0352 6414 ALLCLK,     6414
0353 0000 PASS,       0
0354 7366 M412,      -412
0355 0017 TIME,       17

                        /NO INTERRUPT TIME OUT COUNTER
                        /OCTAL TIME, IN MICROSECONDS, REQUIRED
                        /FOR ONE TIMER LOOP

```

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```

*400
/INTERRUPT ROUTINE - ALL CLOCKS
0400 7300 COMCLK,      CLL CLA
0401 1136 TAD Z TIMOUT      /CHECK CLOCK INTERRUPT STATUS
0402 7640 SZA CLA
0403 5264 JMP INTERR          /JUMP TO ERROR ROUTINE
0404 1007 TAD Z FSTPAS      /IS THIS THE 1ST INTERRUPT?
0405 7640 SZA CLA
0406 5217 JMP NORMAL          /THIS IS NOT THE 1ST INTERRUPT

/FIRST INTERRUPT FROM SPECIFIED CLOCK
0407 1234 TAD M10
0410 3237 DCA INTCNT          /PRESET PASS COUNTER
0411 1241 TAD ATABLE
0412 3240 DCA STORE          /PRESET ADDRESS OF COUNT STORAGE
0413 2007 ISZ Z FSTPAS
0414 1235 TAD P142          /PRESET COUNT BEFORE RE-ENTERING TIMER
0415 3006 DCA Z COUNT
0416 5476 JMP I Z ACLFLG      /RETURN TO INTERRUPT TIMER

/COME HERE FOR NORMAL INTERRUPT PASS
0417 4042 NORMAL,      JMS Z SR1          /SCOPE MODE? (SR1 SET)
0420 5223 JMP .+3          /NO
0421 3006 DCA Z COUNT      /YES; ENABLE AND CLEAR CLOCK FLAG AND
0422 5476 JMP I Z ACLFLG      /RETURN TO TIMER

0423 2240 ISZ STORE
0424 1006 TAD Z COUNT
0425 3640 DCA I STORE      /STORE INTERRUPT TIME INTERVAL
0426 2237 ISZ INTCNT      /IS THIS THE EIGHTH INTERRUPT PASS?
0427 5231 JMP .+2          /NO
0430 5252 JMP LSTPAS      /YES, DISABLE CLOCK FLAG
0431 1236 TAD P161
0432 3006 DCA Z COUNT      /PRESET COUNT BEFORE RE-ENTERING TIMER
0433 5476 JMP I Z ACLFLG      /RETURN TO TIMER

/PG2 - CONSTANTS AND VARIABLES
0434 7770 M10,          -10
0435 0142 P142,          0142
0436 0161 P161,          0161
0437 0000 INTCNT,        0
0440 0000 STORE,         0
0441 0441 ATABLE,        TIMTBL-1

0442 0000 TIMTBL,         0          /INTERRUPT TIME INTERVALS FOR EIGHT
0443 0000 0              /SUCCESSIVE PASSES WILL BE STORED IN
0444 0000 0              /THIS TABLE
0445 0000 0
0446 0000 0
0447 0000 0
0450 0000 0
0451 0000 0

```

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```

/COME HERE FOR EIGHTH INTERRUPT PASS
0452 2136 LSTPAS,      ISZ Z TIMEOUT      /UPDATE INTERRUPT AND TIME-OUT STATUS
0453 1137      TAD Z CLOCK
0454 7106      CLL RTL
0455 7004      RAL
0456 1263      TAD TTXOFF
0457 3260      DCA .+1
0460 6422 DISABL,      TT1OFF      /INSTRUCTION IS MODIFIED FOR SPECIFIED CLOCK
0461 3006      DCA Z COUNT      /RESET INTERRUPT TIMER
0462 5500      JMP I Z AINTON      /RETURN TO TIMER

0463 6412 TTXOFF,      6412

/COME HERE IF INTERRUPT OCCURS AFTER
/CLOCK HAS BEEN DISABLED

0464 4020 INTERR,      JMS Z SR0      /DO WE HALT ON ERROR? (SR.0 SFT)
0465 5260      JMP DISABLE      /NO. DISABLE CLOCK AND CONTINUE
0466 7402 ECT5,      HLT      /INTERRUPT FROM SPECIFIED CLOCK AFTER
                                /CLOCK HAD BEEN DISABLED
0467 5260      JMP DISABLE

/EXIT FROM CLOCK TEST
0470 4051 EXIT,      JMS Z SR2      /DO WE TYPE?
0471 5310      JMP NOTYPE      /NO
                                /TYPE INTERRUPT TIME INTERVALS
0472 1234      TAD M10
0473 3237      DCA INTCNT      /PRESET PASS COUNTER
0474 1241      TAD ATABLE
0475 3240      DCA STORE      /PRESET ADDRESS OF COUNT STORAGE
0476 6046      TLS      /INITIALIZE TTY
0477 4314      JMS CRLF
0500 4332      JMS HEADER
0501 4314      JMS CRLF
0502 2240 GETNXT,      ISZ STORE
0503 1640      TAD I STORE      /GET TIME TO BE TYPED
0504 4763      JMS I ADEC      /CONVERT TIME TO DECIMAL AND TYPE
0505 2237      ISZ INTCNT
0506 5302      JMP GETNXT
0507 4314      JMS CRLF

/ALL TIMES ARE NOW TYPED
0510 4060 NOTYPE,      JMS SR3      /DO WE REPEAT CLOCK TEST? (SR3 RESET)
0511 5477      JMP I ARSTRT      /YES, REPEAT
0512 7402 ECT,      HLT      /END OF 680 CLOCK TEST
0513 5477      JMP I ARSTRT      /DEPRESS CONTINUE TO REPEAT CLOCK TEST

```

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```

0514 0000 /SEND CR AND LF
0515 1322 CRLF, 0
0516 4324 TAD CR
0517 1323 JMS TYPE
0520 4324 TAD LF
0521 5714 JMS TYPE
0522 0215 JMP I CRLF
0523 0212 CR, 215
LF, 212

```

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```

/TRANSMIT SUBROUTINE
0524 0000 TYPE,      0
0525 6041 TSF
0526 5325 JMP .-1
0527 6046 TLS /TRANSMIT CONTENTS OF AC4-AC11
0530 7300 CLA CLL
0531 5724 JMP I TYPE
/TYPE CLOCK NUMBER
0532 0000 HEADER,    0
0533 1352 TAD ALETR
0534 3353 DCA GETLTR
0535 1137 TAD Z CLOCK
0536 1362 TAD ASCI
0537 3361 DCA LETTR+5
0540 1351 TAD M6
0541 3350 DCA FINISH
0542 1753 TAD I GETLTR /PLACE ASCII CHARACTER IN AC
0543 4324 JMS TYPE
0544 2353 ISZ GETLTR
0545 2350 ISZ FINISH /HAS THE LAST CHAR BEEN TYPED
0546 5342 JMP .-4 /NO, TYPE NEXT
0547 5732 JMP I HEADER /YES, RETURN TO MAIN PROGRAM

0550 0000 FINISH,    0

0551 7772 M6,      -6
0552 0554 ALETR,   LETTR
0553 0000 GETLTR,    0

0554 0303 LETTR,    0303 /C
0555 0314 0314 /L
0556 0317 0317 /O
0557 0303 0303 /C
0560 0313 0313 /K
0561 0000 0000 /1,2,3, OR 4

0562 0260 ASCI,    0260
0563 0600 ADEC,    DEC

```

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PAUSE

/680 DCS EXPANDED STATIC TEST - TAPE 2
*600

/OCTAL TO DECIMAL CONVERSION ROUTINE

0600	0000	DEC,	0		
0601	3222	DCA WORK		/STORE OCTAL TIME	
0602	3223	DCA ANSR			
0603	1224	TAD ADRAKA			
0604	3225	DCA PUTBAK			
0605	1226	TAD ASURTR			
0606	3227	DCA AWAY			
0607	4241	JMS CONVRT		/TYPE THOUSANDS	
0610	4241	JMS CONVRT		/TYPE HUNDREDS	
0611	4241	JMS CONVRT		/TYPE TENS	
0612	1222	TAD WORK			
0613	1230	TAD ASCII			
0614	4632	JMS I ATYPE		/TYPE UNITS	
0615	1231	TAD SPACE			
0616	4632	JMS I ATYPE			
0617	1231	TAD SPACE			
0620	4632	JMS I ATYPE		/TYPE TWO SPACES	
0621	5600	JMP I DEC		/GO GET NEXT OCTAL TIME	
0622	0000	WORK,	0		
0623	0000	ANSR,	0		
0624	0633	ADRAKA,	ADRAK		
0625	0000	PUTBAK,	0		
0626	0636	ASURTR,	SURTR		
0627	0000	AWAY,	0		
0630	0260	ASCII,	0260		
0631	0240	SPACE,	0240		
0632	0524	ATYPE,	TYPE		
0633	1750	ADPAK,	1750		/+1000
0634	0144	0144		/+100	
0635	0012	0012		/+12	
0636	6030	SURTR,	6030		/-1000
0637	7634	7634		/-100	
0640	7766	7766		/-10	

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0641	0000	CONVRT,	0	
0642	1222	TAD WORK		
0643	1627	TAD I AWAY		/SUBTRACT 1000, 100, OR 10
0644	7420	SNL		
0645	5251	JMP .+4		
0646	2223	ISZ ANSR		
0647	7100	CLL		
0650	5243	JMP .-5		
0651	1625	TAD I PUTBAK		/ADD BACK 1000, 100, OR 10
0652	3222	DCA WORK		/STORE REMAINDER
0653	1223	TAD ANSR		
0654	1230	TAD ASCII		/CONVERT RESULTS TO ASCI CODE
0655	4632	JMS I ATYPE		
0656	3223	DCA ANSR		/CLEAR ANSR.
0657	2227	ISZ AWAY		
0660	2225	ISZ PUTBAK		
0661	5641	JMP I CONVRT		

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#674

/IOT INSTRUCTION DEFINITIONS

TTCL=6411 /CLEAR LSR TO ZERO. (685)
 TTSL=6412 /INCLUSIVE OR AC5-11 WITH LSR0-6 (685)
 TTRL=6414 /INCLUSIVE OR LSR0-6 WITH AC6-11 (685)
 TTINCR=6401 /INCREMENT LSR BY 1 (681 AND 685)
 TTI=6402 /RECEIVE DATA (681)
 TTU=6404 /TRANSMIT DATA (681)

/681-685 DEVICE SELECTION TEST. (TEST DST)

		/DSTA	
0674	7300	ST685, CLA CLL	/VERIFY THAT DEV. SEL. 41 DOES NOT /SELECT 681
0675	1351	TAD C4000	
0676	3136	DCA Z TIMEOUT	/PRESET INTERRUPT STATUS
0677	6001	ION	
0700	1351	TAD C4000	
0701	6414	TTRL	/SELECT 685. AC0 SHOULDN'T BE AFFECTED
0702	6352	AND C6000	
0703	3140	DCA Z STORAC	/STORE CONTENTS OF AC
0704	4042	JMS Z SR1	/SCOPE REPEAT? (SR1 SET)
0705	5307	JMP .+2	/NO
0706	5274	JMP ST685	/YES
0707	1140	TAD Z STORAC	
0710	1353	TAD M4000	/DOES AC EQUAL 4000?
0711	7440	SZA	
0712	5316	JMP ERDSTA	/NO
0713	4051	JMS Z SR2	/YES, VERIFY REPEAT? (SR2 SET)
0714	5324	JMP DSTH	/NO
0715	5274	JMP ST685	/YES, REPEAT DSTA
		/DSTA FAILED	
0716	4020	ERDSTA, JMS Z SR0	/HALT ON ERROR? (SR0 SET)
0717	5313	JMP .-4	/NO
0720	7100	CLL	
0721	1140	TAD Z STORAC	/YES
0722	7402	ERDST1, HLT	/IOT 6414 SHOULD NOT EFFECT AC0
0723	5274	JMP ST685	/REPEAT DSTA

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0724	7300	/DSTR DSTR,	CLA CLL	/VERIFY THAT DEV. SEL. 40 DOES NOT /SELECT 685
0725	1354	TAD C177		
0726	6412	TTSL		/LOAD ONES INTO LSR
0727	7300	CLA CLL		
0730	6404	ITD		/SHOULD LEAVE AC EQUAL TO ZERO
0731	3140	DCA Z STORAC		/STORE AC
0732	4042	JMS Z SR1		/SCOPE REPEAT? (SR1 SET)
0733	5335	JMP .+2		/NO
0734	5324	JMP DSTR		/YES
0735	1140	TAD Z STORAC		/DOES AC EQUAL ZERO?
0736	7440	SZA		
0737	5343	JMP ERDSTR		/NO
0740	4051	JMS Z SR2		/YES; VERIFY REPEAT? (SR2 SET)
0741	5350	JMP EXDSTR		/NO
0742	5324	JMP DSTR		/REPEAT DSTR
/DSTR FAILED				
0743	4020	ERDSTR,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
0744	5340	JMP .-4		/NO
0745	1140	TAD Z STORAC		/YES
0746	7402	ERDSTR,	HLT	/AC=0 BEFORE IOT 6404 WAS EXECUTED
0747	5324	JMP DSTR		/REPEAT DSTR
0750	5755	EXDSTR,	JMP I ALSRTS	/ENTER LINE SEL. REG. TEST
0751	4000	C4000,	4000	
0752	6000	C6000,	6000	
0753	4000	M4000,	-4000	
0754	0177	C177,	0177	
0755	1000	ALSRTS,	LSRTST	
/RING BELL AFTER 680 TEST CYCLES IF SR5 IS SET				
0756	7604	RING,	CLA OSR	
0757	0374	AND MSKSR5		
0760	7650	SNA CLA		
0761	5274	JMP ST685		/SR5 RESET
0762	2375	ISZ RNGCNT		
0763	1375	TAD RNGCNT		
0764	1376	TAD M1253		
0765	7640	SZA CLA		
0766	5274	JMP ST685		
0767	3375	DCA RNGCNT		
0770	1373	TAD BELCHA		
0771	6046	TLS		/RING BELL
0772	5274	JMP ST685		
0773	0207	BELCHA,	0207	
0774	0100	MSKSR5,	0100	
0775	0000	RNGCNT,	0	
0776	6525	M1253,	-1253	

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*1000

/685 LINE SELECTION REGISTER TEST (LSRTST)

/TEST LSRA

1000	7340	LSRTST,	CLL CLA CMA	/7777 TO AC
1001	6412	TTSL		/AC SHOULD BE CLEARED TO ZERO
1002	3140	DCA Z	STORAC	/STORE AC
1003	4042	JMS Z	SR1	/SCOPE REPEAT? (SR1 SET)
1004	5206	JMP	+.2	/NO
1005	5200	JMP	LSRTST	/YES
1006	1140	TAD Z	STORAC	/DOES AC EQUAL ZERO?
1007	7440	SZA		
1010	5214	JMP	FRLSRA	/NO
1011	4051	JMS Z	SR2	/YES; VERIFY REPEAT? (SR2 SET)
1012	5223	JMP	LSRH	/NO, ENTER NEXT TEST
1013	5200	JMP	LSRTST	/YES, REPEAT TEST LSRA

/TEST LSRA FAILED

1014	4020	FRLSRA,	JMS Z	SR0	/HALT ON ERROR? (SR0 SET)
1015	5211	JMP	.-4	/NO	
1016	1140	TAD Z	STORAC		
1017	7402	FRLSR1,	HLT		/IOT 6412 DID NOT CLEAR AC TO 0
1020	5200	JMP	LSRTST	/REPEAT TEST LSRA	

1021	7777	C7777,	7777
1022	6001	M7777,	-7777

/TEST LSRB

1023	7340	LSRB,	CLL CLA CMA	/7777 TO AC
1024	6411	TTCL		/AC SHOULD NOT BE AFFECTED
1025	3140	DCA Z	STORAC	/STORE AC
1026	4042	JMS Z	SR1	/SCOPE MODE? (SR1 SET)
1027	5231	JMP	+.2	/NO
1030	5223	JMP	LSRB	/YES
1031	1140	TAD Z	STORAC	
1032	1222	TAD	M7777	/DOES AC EQUAL 7777?
1033	7440	SZA		
1034	5240	JMP	FRLSRB	/NO
1035	4051	JMS Z	SR2	/YES; VERIFY REPEAT? (SR2 SET)
1036	5246	JMP	LSRB	/NO, ENTER NEXT TEST
1037	5223	JMP	LSRB	/YES, REPEAT LSRB

/TEST LSRB FAILED

1040	4020	FRLSRB,	JMS Z	SR0	/HALT ON ERROR? (SR0 SET)
1041	5235	JMP	.-4	/NO	
1042	7100	CLL			
1043	1140	TAD Z	STORAC		
1044	7402	FRLSR2,	HLT		/AC = 7777 BEFORE IOT 6411 WAS EXECUTED
1045	5223	JMP	LSRB		

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/TEST LSRC - READ LSR INTO AC, (INCLUSIVE OR)
1046 7240 LSRC,          CLA CMA          /7777 TO AC
1047 6414 TTRL          /AC SHOULD NOT BE AFFECTED
1050 3140 DCA Z STORAC
1051 4042 JMS Z SR1          /SCOPE REPEAT? (SR1 SET)
1052 5254 JMP .+2          /NO
1053 5246 JMP LSRC          /YES
1054 1140 TAD Z STORAC
1055 1222 TAD M7777          /DOES AC EQUAL 7777?
1056 7440 SZA
1057 5263 JMP FRLSRC          /NO
1060 4051 JMS Z SR2          /YES; VERIFY REPEAT? (SR2 SET)
1061 5271 JMP LSRC          /NO, ENTER NEXT TEST
1062 5246 JMP LSRC          /YES, REPEAT TEST LSRC

/TEST LSRC FAILED
1063 4020 FRLSRC,      JMS Z SR0          /HALT ON ERROR? (SR0 SET)
1064 5260 JMP .-4
1065 7100 CLL
1066 1140 TAD Z STORAC
1067 7402 FRLSR3,      HLT          /AC = 7777 BEFORE IOT 6414 WAS EXECUTED
1070 5246 JMP LSRC

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/TEST LSRD - TRY TO LOAD AND READ LSR
1071 7340 LSRD,          CLL CLA CMA          /7777 TO AC
1072 6412 TTSL          /AC5-11 TO LSR, THEN RESET AC
1073 7440 SZA          /WAS AC RESET TO ZERO?
1074 6311 JMP FRACD          /NO
1075 6414 DCNTNU,      TTRL          /YES, NOW READ LSR INTO AC5-11
1076 6140 DCA Z STORAC
1077 4042 JMS Z SR1          /SCOPE REPEAT? (SR1 SET)
1100 6302 JMP .+2          /NO
1101 6271 JMP LSRD          /YES
1102 1140 TAD Z STORAC
1103 1324 TAD M177          /DOES AC EQUAL M177?
1104 7440 SZA
1105 6315 JMP FRLSRD          /NO
1106 4051 DNEXT,        JMS Z SR2          /YES, VERIFY REPEAT? (SR2 SET)
1107 6325 JMP LSRE          /NO, ENTER NEXT TEST
1110 6271 JMP LSRD          /YES, REPEAT TEST LSRD

/TTSL INSTRUCTION DID NOT CLEAR AC
1111 4020 FRACD,        JMS Z SR0          /HALT ON ERROR? (SR0 SET)
1112 6275 JMP DCNTNU          /NO
1113 7402 ERAC1,        HLT          /HALT, NO AC INTERPRETATION
1114 6200 JMP LSRTST          /REPEAT TEST LSRA

/TEST LSRD FAILED
1115 4020 FRLSRD,      JMS Z SR0          /HALT ON ERROR? (SR0 SET)
1116 6306 JMP DNEXT          /NO
1117 7100 CLL
1120 1140 TAD Z STORAC
1121 7402 FRLSR4,      HLT          /AC AND LSR SHOULD EQUAL 177
1122 6271 JMP LSRD          /REPEAT TEST LSRD

1123 0177 01770,        0177
1124 7601 M177,         -0177

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1125 7300 /TEST LSRE - CHECK LSR INCLUSIVE OR LOADING
LSRE, CLA CLL
1126 1323 TAD 01770
1127 6412 TTSL /AC5-11 TO LSR, THEN RESET AC
1130 3141 DCA Z AUXAC /STORE AC
1131 1323 TAD 01770
1132 6412 TTSL
1133 7440 SZA /WAS AC RESET?
1134 5354 JMP ERACE /NO
1135 1141 TAD Z AUXAC /WAS AC RESET BY 1ST TTSL?
1136 7440 SZA
1137 5354 JMP ERACE /NO
1140 6414 ENEXT1, TTSL /AC WAS RESET, NOW READ LSR
1141 3140 DCA Z STORAC /STORE RESULTS OF READ
1142 4042 JMS Z SR1 /SCOPE REPEAT? (SR1 SET)
1143 5345 JMP .+2 /NO
1144 5325 JMP LSRE /YES
1145 1140 TAD Z STORAC
1146 1324 TAD M177 /DOES AC EQUAL 0177?
1147 7440 SZA
1150 5360 JMP FRLSRE /NO
1151 4051 ENEXT2, JMS Z SR2 /YES, VERIFY REPEAT? (SR2 SET)
1152 5766 JMP I ALSRF /NO, ENTER NEXT TEST
1153 5325 JMP LSRE /YES, REPEAT TEST LSRE

/AC NOT RESET BY TTSL
1154 4020 ERACE, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1155 5340 JMP ENEXT1 /NO
1156 7402 ERAC2, HLT
1157 5200 JMP LSRTST /YES, REPEAT LSRA

/TEST LSRE FAILED
1160 4020 ERLSRE, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1161 5351 JMP ENEXT2 /NO
1162 7100 CLL
1163 1140 TAD Z STORAC
1164 7402 FRLSR5, HLT /AC AND LSR SHOULD EQUAL 177
1165 5325 JMP LSRE

1166 1200 ALSRF, LSRF

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#12000

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/TEST LSRE - SUCCESSIVELY LOAD LSR WITH 177 AND 000.
1270 7340 LSRE,      CLL CLA CMA      /7777 TO AC
1271 7412 TTSL      /AC5-11 TO LSR
1272 8141 DCA Z AUXAC      /STORE AC
1273 7412 TTSL      /AC5-11 (0) TO LSR
1274 7440 SZA      /IS AC STILL ZERO
1275 5225 JMP ERACE      /NO
1276 1141 TAD Z AUXAC      /WAS AC RESET BY 1ST TTSL?
1277 7440 SZA
1278 5225 JMP ERACE      /NO
1279 7414 ENEXT1,      TTSL      /AC PROPERLY RESET, NOW READ LSR
1280 8140 DCA Z STORAC      /STORE READ RESULT
1281 4042 JMS Z SR1      /SCOPE REPEAT? (SR1 SET)
1282 5216 JMP .+2      /NO
1283 5200 JMP LSRE      /YES
1284 1140 TAD Z STORAC
1285 1237 TAD M177A      /DOES AC EQUAL 177?
1286 7440 SZA      /SKIP IF EQUAL
1287 5231 JMP ERLSRF      /NO
1288 4051 ENEXT2,      JMS Z SR2      /VERIFY REPEAT? (SR2 SET)
1289 5240 JMP LSRE      /NO
1290 5200 JMP LSRE      /YES

/AC NOT RESET BY TTSL
1291 4020 ERACE,      JMS Z SR0      /HALT ON ERROR? (SR0 SET)
1292 5211 JMP ENEXT1      /NO
1293 7402 ERAC3,      HLT
1294 5766 JMP 1 ALSRA      /REPEAT TEST LSRA

/TEST LSRE FAILED
1295 4020 ERLSRF,      JMS Z SR0      /HALT ON ERROR?
1296 5222 JMP ENEXT2      /NO
1297 7100 CLL
1298 1140 TAD Z STORAC
1299 7402 ERLSR6,      HLT      /AC AND LSR SHOULD EQUAL 177
1300 5200 JMP LSRE      /REPEAT TEST LSRE

1301 7601 M177A,      -177

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1240 7340 /TEST LSRG - TRY TO CLEAR LSR
1241 6412 LSRG,      CLL CLA CMA      /7777 TO AC
1242 7440 TTSL      /AC5-11 TO LSR
1243 5260 SZA      /WAS AC CLEARED TO ZERO?
1244 6411 JMP FRACG /NO
1245 6414 GNEXT1,   TTCL      /RESET LSR TO ZERO
1246 3140 TTPL      /READ CONTENTS OF LSR
1247 4042 DCA Z STORAC /STORE READ RESULTS
1250 5252 JMS Z SR1 /SCOPE REPEAT? (SR1 SET)
1251 5240 JMP .+2     /NO
1252 1140 JMP LSRG    /YES
1253 7440 TAD Z STORAC
1254 5264 SZA      /DOES AC EQUAL ZERO?
1255 4051 JMP ERLSRG  /NO
1256 5271 GNEXT2,   JMS Z SR2      /VERIFY REPEAT? (SR2 SET)
1257 5240 JMP LSRH    /NO
1260 4020 JMS Z SR0
1261 5244 JMP GNEXT1  /NO
1262 7402 FRAC4,    HLT
1263 5766 JMP I ALSRA /REPEAT TEST LSRA

1264 4020 /TEST LSRG FAILED
1265 5255 ERLSRG,   JMS Z SR0      /HALT ON ERROR? (SR0 SET)
1266 1140 JMP GNEXT2  /NO
1267 7402 TAD Z STORAC
1270 5240 ERLSR7,   HLT      /NOT 6411 DID NOT CLEAR LSR
                                /REPEAT TEST LSRG

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/TEST LSRH - SET LSR 0, 2, 4 AND 6 TO ONE

1271	6411	LSRH,	TTCL	/RESET LINE SELECTION REG
1272	7300	CLA CLL		
1273	1325	TAD C125		
1274	6412	TTSL		/AC5-11 TO LSP
1275	7440	SZA		/DID TTSL RESET AC?
1276	6313	JMP ERACH		/NO
1277	6414	HNEXT1,	TTTL	/READ LSR TO AC
1300	3140	DCA Z STORAC		/STORE READ RESULTS
1301	4042	JMS Z SR1		/SCOPE REPEAT? (SR1 SET)
1302	6304	JMP .+2		/NO
1303	6271	JMP LSRH		/YES
1304	1140	TAD Z STORAC		
1305	1326	TAD M125		/DOFS AC EQUAL 0125?
1306	7440	SZA		/SKIP IF EQUAL
1307	6317	JMP ERLSRH		/NO
1310	4051	HNEXT2,	JMS Z SR2	/VERIFY REPEAT? (SR2 SET)
1311	6327	JMP LSR1		/NO
1312	6271	JMP LSRH		/YES
/AC NOT RESET BY TTSL				
1313	4020	ERACH,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1314	6277	JMP HNEXT1		/NO
1315	7402	ERAC5,	HLT	/YES
1316	6766	JMP I ALSRA		/REPEAT TEST LSRA
/TEST LSRH FAILED				
1317	4020	ERLSRH,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1320	6310	JMP HNEXT2		/NO
1321	7100	CLL		
1322	1140	TAD Z STORAC		
1323	7402	ERLSR8,	HLT	/AC AND LSR SHOULD EQUAL 125
1324	6271	JMP LSRH		/REPEAT TEST LSRH
1325	0125	C125,	0125	
1326	7653	M125,	-0125	

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/TEST LSRI - SFT LSRI, 3 AND 5 TO ONE

1327	6411	LSRI,	TTCL	/RESET LSR
1330	7300	CLA CLL		
1331	1363	TAD C052		
1332	6412	TTSL	/AC5-11 TO LSR	
1333	7440	SZA	/DID TTSL RESET AC?	
1334	5351	JMP ERACI	/NO	
1335	6414	INEXT1,	TTCL	/READ LSR TO AC
1336	3140	DCA Z STORAC	/STORE TEST RESULT	
1337	4042	JMS Z SR1	/SCOPE REPEAT? (SR1 SET)	
1340	5342	JMP .+2	/NO	
1341	5327	JMP LSRI	/YES	
1342	1140	TAD Z STORAC		
1343	1364	TAD M052	/DOES AC EQUAL 052?	
1344	7440	SZA		
1345	5355	JMP ERLSRI	/NO	
1346	4051	INEXT2,	JMS Z SR2	/VERIFY REPEAT? (SR2 SET)
1347	5765	JMP I ALSRJ	/NO, ENTER NEXT TEST	
1350	5327	JMP LSRI	/YES, REPEAT TEST LSRI	
/AC NOT RESET BY TTSL				
1351	4020	ERACI,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1352	5335	JMP INEXT1	/NO	
1353	7402	ERAC6,	HLT	
1354	5766	JMP I ALSRA	/REPEAT TEST LSRA	
/TEST LSRI FAILED				
1355	4020	ERLSRI,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1356	5346	JMP INEXT2	/NO	
1357	7100	CLL		
1360	1140	TAD Z STORAC		
1361	7402	ERLSR9,	HLT	/AC AND LSR SHOULD EQUAL 052
1362	5327	JMP LSRI	/REPEAT TEST LSRI	
1363	6052	C052,	0052	
1364	7726	M052,	-0052	
1365	1400	ALSRJ,	LSRJ	
1366	1000	ALSRA,	LSRTST	

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*1400

/TEST LSRJ - CLEAR, LOAD AND READ LSR (IOT 6417)

1400	7300	LSRJ, CLL CLA	
1401	3237	OCA LINCNT	/RESET LINE COUNT
1402	1237	TAD LINCNT	
1403	6417	6417	/CLEAR, LOAD AND READ LSR
1404	3140	OCA Z STORAC	/STORE READ RESULT
1405	4042	JMS Z SR1	/SCOPE REPEAT? (SR1 SET)
1406	5210	JMP .+2	/NO
1407	5202	JMP LSRJ+2	/REPEAT WITH SAME LINE NUMBER
1410	1237	TAD LINCNT	
1411	7041	CIA	/AC EQUALS MINUS LINCNT
1412	1140	TAD Z STORAC	/WAS CORRECT LINE COUNT READ?
1413	7440	SZA	
1414	5225	JMP FRLSRJ	/NO
1415	2237	JNEXT1, ISZ LINCNT	/YES, INCREMENT LINE COUNT
1416	1237	TAD LINCNT	
1417	1240	TAD M200	/HAS LINE COUNT OF 177 BEEN TESTED?
1420	7640	SZA CLA	
1421	5202	JMP LSRJ+2	/NO
1422	4051	JMS Z SR2	/YES; VERIFY REPEAT?
1423	5241	JMP LSRK	/NO, ENTER NEXT TEST
1424	5200	JMP LSRJ	
/TEST LSRJ FAILED			
1425	4020	FRLSRJ, JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1426	5215	JMP JNEXT1	/NO
1427	7100	CLL	
1430	1237	TAD LINCNT	
1431	7402	FRLSR10, HLT	/HALT WITH CORRECT LINE COUNT IN AC
1432	7200	CLA	
1433	1140	TAD Z STORAC	
1434	7402	HLT	/HALT WITH INCORRECT TEST RESULT IN AC
1435	7200	CLA	
1436	5202	JMP LSRJ+2	
1437	0000	LINCNT, 0	
1440	7600	M200, -0200	

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/TEST LSRK - INCREMENT LSR FROM 000 TO 177

1441	7300	LSRK,	CLL CLA	
1442	3237	DCA LINCNT		/CLEAR LINE COUNT
1443	6411	TTCL		/CLEAR LSR TO ZERO
1444	6401	ADD1,	TTINCR	/INCREMENT LSR BY ONE
1445	4042	JMS $\neq$ SR1		/SCOPE REPEAT? (SR1 SET)
1446	5250	JMP .+2		/NO
1447	5271	JMP AGAIN		/YES
1450	2237	IS $\neq$ LINCNT		
1451	1237	TAD LINCNT		
1452	0313	AND C177K		
1453	3237	DCA LINCNT		
1454	6414	TTRL		/READ LSR TO AC
1455	3140	DCA $\neq$ STORAC		/STORE AC
1456	1237	TAD LINCNT		
1457	7041	CIA		/AC CONTAINS - LINE COUNT
1460	1140	TAD $\neq$ STORAC		
1461	7440	SZA		/DID LSR INCREMENT CORRECTLY?
1462	5274	JMP ERLSRK		/NO

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1463	1237	KNEXT,	TAD LINCNT	/YES
1464	7440	SZA		/DOFS LINE COUNT EQUAL 000
1465	5244	JMP ADD1		/NO
1466	4051	JMS Z SR2		/YES; VERIFY REPEAT? (SR2 SET)
1467	5312	JMP FXLSR		/NO
1470	5241	JMP LSRK		/YES, REPEAT LSRK
/SET UP SCOPE REPEAT				
1471	1237	AGAIN,	TAD LINCNT	
1472	6413	6413		/CLEAR LSR AND LOAD WITH CURRENT
1473	5244	JMP ADD1		/LINE COUNT
/LSR DID NOT INCREMENT CORRECTLY				
1474	4020	FLSRK,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1475	5263	JMP KNEXT		/NO
1476	7100	CLL		
1477	1237	TAD LINCNT		
1500	7402	FLSR11,	HLT	/HALT WITH CORRECT LINE COUNT IN AC
1501	7200	CLA		
1502	1140	TAD X STORAC		
1503	7402	HLT		/HALT WITH INCORRECT LINE COUNT
1504	7200	CLA		/IN AC
1505	1237	TAD LINCNT		
1506	1314	TAD M1		/SUBTRACT 1 FROM LINE COUNT
1507	1313	AND C177K		
1510	7237	DCA LINCNT		/RETEST SAME LINE COUNT
1511	5271	JMP AGAIN		
1512	5715	FXLSR,	JMP I ATS681	/ENTER 681 INSTRUCTION TESTS
1513	0177	C177K,	0177	
1514	7777	M1,	-0001	
1515	1600	ATS681,	TS681A	

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PAUSE

/680 DCS EXPANDED STATIC TEST - TAPE 3
/681 INSTRUCTION TEST

*1600
/TS681A - 681 NO ACTIVITY TEST (IOT 6400)

1600	7200	TS681A, CLA	
1601	1237	TAD OKJMP1	
1602	3207	DCA NOAC+1	
1603	1240	TAD FRJMP1	
1604	3210	DCA NOAC+2	/PRESET JMP INSTRUCTIONS
1605	7120	CLL CML	/1 TO LINK
1606	6400	NOAC, 6400	/ADDRESS 681 BUT DEFINE NO ACTION
1607	5212	JMP NOSTAT	/THIS INSTRUCTION SHOULD BE EXECUTED
1610	5223	JMP SKIPER	/DON'T EXECUTE
1611	5227	JMP TTIER	/DON'T EXECUTE
1612	4042	NOSTAT, JMS Z SR1	/SCOPE REPEAT? (SR1 SET)
1613	5215	JMP .+2	/NO
1614	5200	JMP TS681A	/YES
1615	7430	SZL	/WAS THE LINK RESET?
1616	5220	JMP .+2	/NO
1617	5233	JMP TTIER	/YES
1620	4051	JMS Z SR2	/VERIFY REPEAT? (SR2 SET)
1621	5241	JMP TS681B	/NO, ENTER NEXT TEST
1622	5200	JMP TS681A	/YES, REPEAT TS681A
/INSTRUCTION FOLLOWING 6400 WAS NOT EXECUTED			
1623	4020	SKIPER, JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1624	5212	JMP NOSTAT	/NO
1625	7402	FRIT01, HLT	/HALT 1 - INCORRECT 681 SKIP ENABLE
1626	5200	JMP TS681A	/REPEAT TEST
/TWO INSTRUCTIONS FOLLOWING 6400 WERE NOT EXECUTED			
1627	4020	TTIER, JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1630	5212	JMP NOSTAT	/NO
1631	7402	FRIT02, HLT	/HALT 2 - IOT 6400 INTERPRETED AS IOT 6402
1632	5200	JMP TS681A	/REPEAT TEST
/LINK BIT WAS RESET			
1633	4020	TTIER, JMS Z SR0	/HALT ON ERROR? (SR0 SET)
1634	5220	JMP SKIPER-3	/NO
1635	7402	FRIT03, HLT	/HALT 3 - IOT 6400 INTERPRETED AS IOT 6404
1636	5200	JMP TS681A	/REPEAT TEST
1637	5212	OKJMP1, JMP NOSTAT	
1640	5223	FRJMP1, JMP SKIPER	

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/TS681B - TIO TEST, (6404), PART 1 - CHECK ZERO THE LINK
1641 7320 TS681B, CLA CLL CML /0 TO AC, 1 TO LINK
1642 6404 TIO /CONTENTS OF LINE SELECTION REGISTER
/SHOULD NOT AFFECT THIS TEST
1643 3140 DCA Z STORAC /STORE AC
1644 4042 JMS Z SR1 /SCOPE REPEAT? (SR1 SET)
1645 5247 JMP .+2 /NO
1646 5241 JMP TS681B /YES
1647 7430 SZL /WAS LINK RESET TO ZERO?
1648 5260 JMP LN01Z /NO
1649 1140 TAD Z STORAC /YES; NOW EXAMINE AC0
1650 7500 SMA /SKIP IF AC0 (1)
1651 5255 JMP .+2
1652 5265 JMP AC0SET /LINK BIT WAS SHIFTED

1653 4051 PT1NXT, JMS Z SR2 /TEST OK; VERIFY REPEAT? (SR2 SET)
1654 5272 JMP TS681C /NO, ENTER TEST PART 2
1655 5241 JMP TS681B /REPEAT PART 1

/LINK BIT WAS NOT RESET.
1656 4020 LN01Z, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1657 5255 JMP PT1NXT /NO
1658 1140 TAD Z STORAC /HALT AND DISPLAY CONTENTS OF
1659 7402 ERIT04, HLT /AC FOLLOWING TIO EXECUTION
1660 5241 JMP TS681B /REPEAT TIO TEST, PART 1

/AC0 WAS SET TO ONE.
1661 4020 AC0SET, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1662 5255 JMP PT1NXT /NO
1663 1140 TAD Z STORAC /HALT AND DISPLAY CONTENTS OF AC
1664 7402 ERIT05, HLT /FOLLOWING TIO EXECUTION
1665 5241 JMP TS681B /REPEAT TIO TEST, PART 1

/TS681C - TIO TEST, PART 2 - CHECK LINK ROT. DISABLE AND RAR
1666 7320 TS681C, CLA CLL CML /1 TO LINK
1667 1325 TAD 04001 /1 TO AC0 AND AC11
1668 6404 TIO /0 TO LINK AND RAR
1669 3140 DCA Z STORAC /STORE AC
1670 4042 JMS Z SR1 /SCOPE REPEAT? (SR1 SET)
1671 5301 JMP .+2 /NO
1672 5272 JMP TS681C /YES
1673 7430 SZL /IS LINK RESET?
1674 5312 JMP LR0TER /NO
1675 1140 TAD Z STORAC /YES
1676 1326 TAD M2000 /WAS AC EQUAL TO 2000 AFTER
1677 7440 SZL /EXECUTION OF TIO?
1678 5317 JMP SHFIER /NO
1679 4051 PT2NXT, JMS Z SR2 /VERIFY REPEAT? (SR2 SET)
1680 5327 JMP TS681D /NO, ENTER TEST PART 3
1681 5272 JMP TS681C /YES, REPEAT PART 2

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1712 4020 /LINK ROTATE NOT DISABLED
1713 5307 LROTFR, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1714 1140 JMP PT2NXT /NO
1715 7402 TAD Z STORAC
1716 5272 ERTT06, HLT /HALT AND DISPLAY AC TEST RESULTS
1716 5272 JMP TS681C /REPEAT PART 2

1717 4020 /RAR NOT EXECUTED CORRECTLY
1720 5307 SHFTFR, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1721 7100 JMP PT2NXT /NO
1722 1140 CLL
1723 7402 TAD Z STORAC
1724 5272 ERTT07, HLT /HALT AND DISPLAY AC TEST RESULTS
1724 5272 JMP TS681C /REPEAT PART 2

1725 4001 C4001, 4001
1726 6000 M2000, -2000

1727 7300 /TS681 - TIO TEST, PART 3
1730 1362 TS681D, CLA CLL
1731 6404 TAD C5252
1732 3140 TIO /TEST RAR EXECUTION
1733 4042 DCA Z STORAC /STORE AC
1734 5336 JMS Z SR1 /SCOPE REPEAT? (SR1 SET)
1735 5327 JMP .+2 /NO
1736 7430 JMP TS681D /YES
1737 5347 SZL /IS LINK STILL RESET?
1740 1140 JMP ERMCL /LINK WAS COMPLIMENTED
1741 1363 TAD Z STORAC /LINK IS OK
1742 7440 TAD M2525 /WAS AC SHIFTED CORRECTLY
1743 5354 SZA
1744 4051 JMP ERSHFT /NO
1745 5764 PT3NXT, JMS Z SR2 /VERIFY REPEAT? (SR2 SET)
1746 5327 JMP I ATTITS /NO, ENTER NEXT TEST
1746 5327 JMP TS681D /YES, REPEAT PART 3

1747 4020 /LINK BIT WAS SET TO ONE.
1750 5344 ERMCL, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1751 1140 JMP PT3NXT /NO
1752 7402 ERTT08, TAD Z STORAC /CHECK FOR INCORRECT CM
1753 5327 HLT
1753 5327 JMP TS681D

1754 4020 /RAR WAS NOT EXECUTED CORRECTLY.
1755 5344 ERSHFT, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
1756 7100 JMP PT3NXT /NO
1757 1140 CLL
1760 7402 TAD Z STORAC
1761 5327 ERTT09, HLT /AC SHOULD EQUAL 2525
1761 5327 JMP TS681D

1762 5252 C5252, 5252
1763 5253 M2525, -2525
1764 2000 ATTITS, TS681E

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/681 INSTRUCTION TEST, (CONTINUED)

*2000

/TS681F - TTI TEST, PART 1 - CHECK IOT 6402 EXECUTION FORMAT

TS681E, CLL CLA CMA

TTSL

/PRESET LINE SELECTION REG. TO 177

TAD SWJMP1

/AND CLEAR AC

DCA TTYIN+1

TAD CWJMP1

DCA TTYIN+2

/PRESET ERROR JUMP INSTRUCTIONS

TTYIN, TTI

/DON'T EXECUTE NEXT TWO WORDS

JMP FRJMP2

/LINE STATUS WORD (LSW)

JMP FRJMP3

/CHARACTER ASSEMBLY WORD (CAW)

JMP OKJMP2

JMP FRJMP4

/681 CORRECTLY SKIPPED LSW AND CAW

OKJMP2, JMS Z SR1

/SCOPE REPEAT? (SR1 SET)

JMP TS681F

/NO, ENTER NEXT TEST

JMP TS681F+2

/YES

/LSW WAS INTERPRETED AS AN INSTRUCTION

FRJMP2, JMS Z SR0

/HALT ON ERROR? (SR0 SET)

JMP OKJMP2

/NO

ERTTI1, HLT

/HALT 1 - CHECK GENERATION OF S CYCLE

JMP TS681E+2

/CAW WAS INTERPRETED AS AN INSTRUCTION

FRJMP3, JMS Z SR0

/HALT ON ERROR?

JMP OKJMP2

/NO

ERTTI2, HLT

/HALT 2 - CHECK S CYCLE 'SKIP BUS IN' ENABLE

JMP TS681E+2

/681 SKIPPED TWICE

FRJMP4, JMS Z SR0

/HALT ON ERROR?

JMP OKJMP2

/NO

ERTTI3, HLT

/HALT 3 - DOUBLE SKIP OCCURRED

JMP TS681E+2

SWJMP1, JMP FRJMP2

CWJMP1, JMP FRJMP3

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/TS681F - TTI TEST, PART 2 - CHECK S CYCLE OPERATIONS
2034 7300 TS681F,      CLL CLA
2035 1331 TAD SW2000
2036 3242 DCA TTYIN2+1
2037 1331 TAD SW2000
2040 3243 DCA TTYIN2+2      /PRESET LSW AND CAW

2041 6402 TTYIN2,      TTI
2042 2000 2000          /LSW
2043 2000 2000          /CAW

2044 4042 JMS Z SR1      /SCOPE REPEAT? (SR1 SET)
2045 5247 JMP .+2        /NO
2046 5234 JMP TS681F     /YES
2047 1242 TAD TTYIN2+1
2050 1337 TAD M1000      /LSW SHOULD CONTAIN 1000
2051 7440 SZA
2052 5262 JMP LSWERR
2053 1243 TAD TTYIN2+2
2054 1336 TAD MN2000     /CAW SHOULD CONTAIN 2000
2055 7440 SZA
2056 5323 JMP CAWERR
2057 4051 NXTTY,      JMS Z SR2      /VERIFY REPEAT? (SR2 SET)
2060 5735 JMP I AT681G /NO, ENTER NEXT TEST
2061 5234 JMP TS681F     /YES

/DETERMINE LSW ERROR TYPE
2062 4020 LSWERR,      JMS Z SR0      /HALT ON ERROR? (SR0 SET)
2063 5257 JMP NXTTY          /NO
2064 1242 TAD TTYIN2+1      /WAS MB SHIFTED?
2065 0331 AND SW2000
2066 7640 SZA CLA          /SKIP IF MB WAS SHIFTED
2067 5271 JMP .+2
2070 5275 JMP .+5

/ERROR HALT 1 - MB WAS NOT SHIFTED CORRECTLY
2071 1242 TAD TTYIN2+1      /DISPLAY LSW IN AC
2072 7100 CLL
2073 7402 ERTTI4,      HLT          /HALT 1- CHECK 'SHIFT MB'
2074 5234 JMP TS681F

/VERIFY CORRECT MB0 (0) SHIFT ENABLED
2075 1242 TAD TTYIN2+1      /WAS MB0 INCORRECTLY SET TO 1?
2076 0332 AND SW5000
2077 1333 TAD M5000
2100 7640 SZA CLA          /SKIP IF MB0 (1) AND THE MB REG.
2101 5306 JMP .+5          /SHIFT WAS EXECUTED

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/ERROR HALT 2
2112 1242 TAD TTYIN2+1 /DISPLAY LSW IN AC
2113 7100 CLL
2114 7402 ERITI5, HLT /HALT 2 - MR0 INCORRECTLY SET TO 1
2115 5234 JMP TS681F

/VERIFY THAT COUNT MR WAS NOT ENABLED
2116 1242 TAD TTYIN2+1 /WAS COUNT MR INCORRECTLY ENABLED?
2117 5334 AND SW0007
2118 7640 SZA CLA /DON'T SKIP IF COUNT WAS ENABLED
2119 5313 JMP .+2
2120 5317 JMP .+5 /LSW ERROR UNIDENTIFIED

/ERROR HALT 3
2113 1242 TAD TTYIN2+1 /DISPLAY LSW IN AC
2114 7100 CLL
2115 7402 ERITI6, HLT /HALT 3 - 'COUNT MR' INCORRECTLY ENABLED
2116 5234 JMP TS681F

/ERROR HALT 4
2117 1242 TAD TTYIN2+1 /DISPLAY LSW IN AC
2118 7100 CLL
2119 7402 ERITI7, HLT /HALT 4 - COMBINATION OF ERRORS 4,5 & 6
2120 5234 JMP TS681F

/ERROR HALT 5 - CAW WAS MODIFIED
2123 4020 CAWERR, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
2124 5257 JMP NXYTY /NO
2125 1243 TAD TTYIN2+2 /DISPLAY CAW IN AC
2126 7100 CLL
2127 7402 ERITI8, HLT /HALT 5 - CHECK FOR INCORRECT C CYC
2128 5234 JMP TS681F

2131 2000 SW2000, 2000
2132 5000 SW5000, 5000
2133 3000 M5000, -5000
2134 0007 SW0007, 0007
2135 2200 AT681G, TS681G
2136 6000 MN2000, -2000
2137 7000 M1000, -1000

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*2200

/TS681G - TTI TEST, PART 3 - CHECK IOT 6402 WITH LSW = 4003

2200	7300	TS681G,	CLL CLA	
2201	1254	TAD LS4003		/PRESET LINE STATUS WORD
2202	3210	DCA TTYIN3+1		
2203	1256	TAD CW4000		/PRESET CHARACTER ASSEMBLY WORD
2204	3211	DCA TTYIN3+2		
2205	1260	TAD ANOP		
2206	3212	DCA TTYIN3+3		/PRESET NOP INSTRUCTION
2207	6402	TTYIN3,	TTI	
2210	4003	4003		/LSW
2211	4000	4000		/CAW
2212	7000	NOP		
2213	4042	JMS Z SR1		/SCOPE REPEAT? (SR 1 SET)
2214	5216	JMP .+2		/NO
2215	5200	JMP TS681G		/YES
2216	1210	TAD TTYIN3+1		
2217	1255	TAD M4004		/WAS LSW UPDATED CORRECTLY?
2220	7440	SZA		
2221	5231	JMP FRRLSW		/LSW IS INCORRECT
2222	1211	TAD TTYIN3+2		
2223	1257	TAD M6000		/WAS CAW UPDATED CORRECTLY?
2224	7440	SZA		
2225	5236	JMP FRRCAW		/CAW IS INCORRECT
2226	4051	TTYNXT,	JMS Z SR2	/VERIFY REPEAT? (SR2 SET)
2227	5262	JMP TS681H		/NO
2230	5200	JMP TS681G		/YES
		/ERROR HALT 1 - LSW WAS NOT UPDATED CORRECTLY		
2231	4020	FRRLSW,	JMS Z SR0	/HALT ON ERROR? (SR0 SET)
2232	5226	JMP TTYNXT		/NO
2233	1210	TAD TTYIN3+1		/DISPLAY LSW IN AC
2234	7402	ERTTI9,	HLT	/HALT 1 - AC SHOULD EQUAL 4004
2235	5200	JMP TS681G		

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/ERROR HALT 2 - CAW WAS NOT UPDATED CORRECTLY
2236 4020 FRRCAW,      JMS Z SR0                      /HALT ON ERROR? (SR0 SET)
2237 5226      JMP TTYNXT                      /NO
2240 1212      TAD TTYIN3+3                    /WAS NOP MODIFIED?
2241 1261      TAD MNOP
2242 7640      SZA CLA
2243 5250      JMP .+5                          /NOP WAS MODIFIED
2244 7100      CLL
2245 1211      TAD TTYIN3+2
2246 7402 ETTI10,      HLT                      /HALT 2A - CAW & AC SHOULD = 6000
2247 5200      JMP TS681G
2250 7100      CLL
2251 1211      TAD TTYIN3+2
2252 7402 ETTI11,      HLT                      /HALT 2B - CHECK FOR INCORRECT S CYCLE
/  SKIP ENABLE
2253 5200      JMP TS681G

2254 4003      LS4003,      4003
2255 3774      M4004,      -4004
2256 4000      CW4000,      4000
2257 2000      M6000,      -6000
2258 7000      ANOP,      NOP
2261 1000      MNOP,      -7000

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/TS681H - TTI TEST, PART 4 - CHECK 'COUNT MB' EXECUTION

2262	7300	TS681H,	CLL CLA	
2263	1366	TAD MSEVEN		/PRESET TTICNT TO -7
2264	3365	DCA TTICNT		
2265	1131	TAD Z FOUR		/PRESET LSWCNT TO 4
2266	3364	DCA LSWCNT		
2267	1256	PASTRT,	TAD CW4000	
2270	1364	TAD LSWCNT		
2271	3301	DCA TTYIN4+1		/PRESET LSW
2272	2364	ISZ LSWCNT		/ADD ONE TO LSWCNT
2273	1364	TAD LSWCNT		
2274	4367	AND MSKCNT		/MSKCNT EQUALS 0007
2275	3364	DCA LSWCNT		
2276	1256	TAD CW4000		/PRESET CAW
2277	3302	DCA TTYIN4+2		
2300	4402	TTYIN4,	TTI	
2301	4004	4004		/LSW - COUNT IS INCREMENTED ON EACH PASS
2302	4000	4000		/CAW - SHOULD NEVER BE MODIFIED
2303	4042	JMS Z SR1		/SCOPE REPEAT? (SR1 SET)
2304	5312	JMP .+6		/NO
2305	1364	RETEST,	TAD LSWCNT	/YES
2306	1371	TAD CNTM1		/SUBTRACT 1 FROM LSWCNT
2307	4367	AND MSKCNT		
2310	3364	DCA LSWCNT		
2311	5267	JMP PASTRT		/EXECUTE REPEAT
		/WAS LSW UPDATED CORRECTLY?		
2312	1301	TAD TTYIN4+1		
2313	7500	SMA		/DOES LSW0 CONTAIN 1?
2314	5334	JMP FRLSW0		/NO; ERROR 1
2315	4367	AND MSKCNT		/YES; CONTINUE TEST
2316	7041	CIA		
2317	1364	TAD LSWCNT		/WAS LSW COUNT INCREMENTED CORRECTLY?
2320	7440	SZA		
2321	5343	JMP ERRCNT		/NO; ERROR 2
2322	1302	TAD TTYIN4+2		/YES; TEST CAW
2323	7041	CIA		
2324	1256	TAD CW4000		/DOES CAW EQUAL 4000?
2325	7440	SZA		
2326	5355	JMP MODCAW		/NO
		/TEST FOR SEVENTH PASS		
2327	2365	CHKPAS,	ISZ TTICNT	
2330	5267	JMP PASTRT		/NOT SEVENTH PASS
2331	4051	JMS Z SR2		/VERIFY REPEAT? (SR2 SET)
2332	5770	JMP I AT681I		
2333	5262	JMP TS681H		/YES, REPEAT TEST

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2334 4020 /ERROR 1 - LSW WAS CLEARED TO ZERO
2335 5327 FRI SW0, JMS Z SR0 /HALT ON ERROR? (SR0 SET)
2336 7100 JMP CHKPAS /NO
2337 1301 CLL
2338 1301 TAD TTYIN4+1 /DISPLAY LSW IN AC
2339 7402 FTTI12, HLT /HALT 1 - CHECK FOR INCORRECT S CYCLE MH SHIFT
2340 7200 CLA
2341 5305 JMP RETEST

2343 4020 /ERROR 2 - LSW COUNT IS INCORRECT
2344 5327 FRCNT, JMS Z SR0 /HALT ON ERROR?
2345 7100 JMP CHKPAS /NO
2346 1364 CLL
2347 1364 TAD LSWCNT /DISPLAY EXPECTED LSW COUNT
2348 7402 FTTI13, HLT /HALT 2A - CHECK S CYCLE 'COUNT MH ENABLE.'
2349 7200 CLA
2350 1301 TAD TTYIN4+1 /DISPLAY ACTUAL LSW COUNT
2351 7402 FTTI14, HLT /HALT 2B
2352 7200 CLA
2353 5305 JMP RETEST

2355 4020 /ERROR 3 - CAW IS INCORRECT
2356 5327 MODCAW, JMS Z SR0 /HALT ON ERROR?
2357 7100 JMP CHKPAS /NO
2358 1302 CLL
2359 1302 TAD TTYIN4+2 /DISPLAY CAW IN AC
2360 7402 FTTI14, HLT /HALT 3 - CHECK FOR INCORRECT C CYCLE GENERATIO
2361 7200 CLA
2362 5305 JMP RETEST

2364 0000 LSWCNT, 0
2365 0000 TTICNT, 0
2366 7771 MSFEVEN, -0007
2367 0007 MSKCNT, 0007
2370 2400 AT681I, TS681I
2371 7777 CNTM1, -0001

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*2400

/TS681I - MB REGISTER SHIFT TEST, PART 1

2400	7300	TS681I,	CLA CLL	/FILL MB WITH ONES
2401	1255	TAD M14		
2402	3252	DCA SHFCNT		/PRESET SHIFT COUNTER
2403	3253	DCA CHKCAW		/RESET TEST WORD
2404	3211	SHIFTI,	DCA TTYIN5+2	/PRESET CAW
2405	1254	TAD PRLSW1		
2406	3210	DCA TTYIN5+1		/PRESET LINE STATUS WORD
2407	6402	TTYIN5,	TTI	
2410	4003	4003		/LSW
2411	0000	%		/CAW
2412	4042	JMS Z SR1		/SCOPE REPEAT?
2413	5216	JMP .+3		/NO
2414	1253	TAD CHKCAW		/YES, RETAIN SAME CAW & REPEAT
2415	5204	JMP SHIFTI		
		/CHECK RESULTS OF MB SHIFT		
2416	7120	CLL CML		/1 TO LINC
2417	1253	TAD CHKCAW		
2420	7010	RAR		/SIMULATE MR SHIFT
2421	3253	DCA CHKCAW		/STORE SHIFT RESULTS
2422	1253	TAD CHKCAW		
2423	7041	CIA		
2424	1211	TAD TTYIN5+2		/COMPARE FOR CORRECT SHIFT RESULTS
2425	7640	SZA CLA		/SKIP IF RESULTS ARE CORRECT
2426	5234	JMP FRMSI		
		/MR SHIFT CORRECT		
2427	2252	SHFINI,	ISZ SHFCNT	/SKIP IF MR IS FULLY, (ONES)
2430	5205	JMP SHIFTI+1		
2431	4051	JMS Z SR2		/VERIFY REPEAT?
2432	5256	JMP TS681J		/NO, ENTER PART 2
2433	5200	JMP TS681I		/YES, REPEAT PART 1

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2434 4020 /MB SHIFT ERROR DETECTED, (PART 1)
2435 5227 ERRBSI, JMS Z SR0 /HALT ON ERROR?
2436 7100 JMP SHEINI /NO
2437 1253 CLL
2438 7402 /HALT 1 - MB SHIFT ERROR, PART 1
2439 1253 TAD CHKCAW
2440 7402 ERMBR1, HLT /HALT & DISPLAY CORRECT SHIFT RESULTS
2441 7200 CLA
2442 1211 TAD TTYIN5+2
2443 7402 HLT /HALT & DISPLAY ACTUAL SHIFT RESULTS

2444 7300 /RETEST FAILING SHIFT
2445 1253 CLA CLL
2446 7004 TAD CHKCAW
2447 3253 RAL /RESTORE PREVIOUS CAW CONTENTS
2448 1253 DCA CHKCAW /AND RETEST
2449 1253 TAD CHKCAW
2450 5204 JMP SHFTI

2452 0000 SHECNT, 0
2453 0000 CHKCAW, 0
2454 4003 PRLSW1, 4003
2455 7764 M14, -0014

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/TS681J - MB REGISTER SHIFT TEST, PART 2

2456	7201	TS681J,	CLA IAC	/FILL MR WITH ZEROS
2457	1255	TAD M14		
2460	3252	DCA SHFCNT		/PRESET SHIFT COUNTER
2461	1331	TAD MS3777		
2462	3253	DCA CHKCAW		/PRESET TEST WORD
2463	1254	SHIFTJ,	TAD PRLSW1	
2464	3270	DCA TTYIN6+1		/PRESET LINE STATUS WORD
2465	1253	TAD CHKCAW		
2466	3271	DCA TTYIN6+2		/PRESET CAW
2467	6402	TTYIN6,	TTI	
2470	4003	4003		/LSW
2471	3777	3777		/CAW
2472	4042	JMS Z SR1		/SCOPE REPEAT?
2473	5275	JMP .+2	/NO	
2474	5263	JMP SHIFTJ		/YES, RETAIN SAME CAW & REPEAT
/CHECK RESULTS OF MB SHIFT				
2475	1253	TAD CHKCAW		
2476	7110	CLL RAR		/SIMULATE MR SHIFT
2477	3253	DCA CHKCAW		
2500	1271	TAD TTYIN6+2		/GET ACTUAL MB SHIFT RESULT
2501	0331	AND MS3777		
2502	7041	CIA		
2503	1253	TAD CHKCAW		/COMPARE WITH CORRECT MB SHIFT RESULT
2504	7640	SZA CLA		/SKIP IF RESULTS ARE CORRECT
2505	5313	JMP FRMBSJ		
/MR SHIFT CORRECT				
2506	2252	SHFINJ,	ISZ SHFCNT	/SKIP IF MR IS FULL, (ZEROS)
2507	5263	JMP SHIFTJ		
2510	4051	JMS Z SR2		/VERIFY REPEAT?
2511	5732	JMP I AT681K	/NO, ENTER PART 3	
2512	5256	JMP TS681J		/YES, REPEAT PART 2

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2513 4020 /MB SHIFT ERROR DETECTED, (PART 2)
      FRMSJ,      JMS Z SR0      /HALT ON ERROR?
2514 5306      JMP SHEINJ
      /HALT 2 - MB SHIFT ERROR, PART 2
2515 7100      CLL
2516 1253      TAD CHKCAW
2517 7402 FRMS2,      HLT      /HALT & DISPLAY CORRECT SHIFT RESULTS
2520 7200      CLA
2521 1271      TAD TTYIN6+2
2522 0331      AND MS3777
2523 7402      HLT      /HALT & DISPLAY ACTUAL SHIFT RESULTS
2524 7200      CLA
2525 1253      TAD CHKCAW
2526 7124      CLL CML HAL      /RESTORE PREVIOUS CAW CONTENTS
2527 3253      DCA CHKCAW      /AND RETET
2530 5263      JMP SHIF TJ
2531 3777 MS3777,      3777
2532 2600 AT681K,      TS681K

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*2600

/TS681K - MB REGISTER SHIFT TEST, PART 3

2600	7300	TS681K,	CLA CLL	/FILL MR WITH ALTERNATE ONES & ZEROS
2601	3302	DCA PSELEC		/RESET PASS SELECTOR
2602	1301	TAD MIN14		
2603	3277	DCA CNTSHF		/PRESET SHIFT COUNTER
2604	1303	TAD PROAW		
2605	3276	DCA CAWCHK		/PRESET TEST WORD
2606	1300	SHIFTK,	TAD PRLSW3	
2607	3213	DCA TTYIN7+1		/PRESET LSW
2610	1276	TAD CAWCHK		
2611	3214	DCA TTYIN7+2		/PRESET CAW TO 4000
2612	6402	TTYIN7,	TTI	
2613	4003	4003		/LSW
2614	4000	4000		/CAW
2615	4042	JMS Z SR1		/SCOPE REPEAT?
2616	5226	JMP .+2	/NO	
2617	5206	JMP SHIFTK		/YES, RETAIN SAME CAW & RETEST
/CHECK RESULTS OF MB SHIFT				
2620	1276	TAD CAWCHK		
2621	7130	CLL CML RAR		/SIMULATE MB SHIFT
2622	3276	DCA CAWCHK		
2623	7010	RAR		
2624	3304	DCA RITBUK		/SAVE PREVIOUS CONTENTS OF MR11
2625	1214	TAD TTYIN7+2		
2626	7041	CIA		
2627	1276	TAD CAWCHK		/COMPARE MB SHIFT RESULTS
2630	7640	SZA CLA		/SKIP IF RESULTS ARE CORRECT
2631	5250	JMP ERMBSK		
/MR SHIFT CORRECT				
2632	2277	SHFINK,	ISZ CNTSHF	/SKIP IF MR FULL, (ONES & ZEROS)
2633	5235	JMP .+2		
2634	5267	JMP EX681T		
2635	1302	TAD PSELEC		/EXAMINE PASS SELECTOR
2636	7640	SZA CLA		
2637	5246	JMP .+7		
2640	1276	TAD CAWCHK		/PASS SELECTOR = 0
2641	1305	AND RSTMR0		/NOW SHIFT MR0 (0) TO MB1
2642	3276	DCA CAWCHK		
2643	7001	IAC		
2644	3302	DCA PSELEC		/1 TO PASS SELECTOR
2645	5206	JMP SHIFTK		

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2616 3302 /PASS SELECTOR = 1
      617 5206 DCA PSFLEC /NO TO PASS SELECTOR
      617 5206 JMP SHIFTK /NOW SHIFT MB1 (1) TO MB1

      /MB SHIFT ERROR DETECTED, (PART 3)
2620 4020 ERMRSK, JMS Z SR1 /HALT ON ERROR?
2621 5232 JMP SHFINK /NO

      /HALT 3 - MB SHIFT ERROR, PART 3
2622 7100 CLL
2623 1276 TAD CAWCHK
2624 7402 ERMR33, HLT /HALT & DISPLAY CORRECT SHIFT RESULTS
2625 7200 CLA
2626 1214 TAD TTYIN7+2
2627 7402 HLT /HALT & DISPLAY ACTUAL SHIFT RESULTS
2628 7300 CLA CLL
2629 1304 TAD RITBUK /GET PREVIOUS CONTENTS OF MB11
2630 7004 RAL
2631 1276 TAD CAWCHK
2632 7004 RAL /RESTORE PREVIOUS CAW CONTENTS
2633 3276 DCA CAWCHK
2634 5206 JMP SHIFTK /RETEST FAILING MB SHIFT

2637 4051 FX681T, JMS Z SR2 /VERIFY REPEAT?
2638 5272 JMP .+2 /NO
2639 5200 JMP TS681K

      /END OF STATIC TEST
2642 4060 JMS Z SR3 /HALT AT COMPLETION OF STATIC TEST?
2643 5706 JMP I ARING /NO, REPEAT STATIC TEST
2644 7402 ENDTST, HLT /END OF STATIC TEST
2645 5706 JMP I ARING /DEPRESS CONTINUE TO REPEAT STATIC TEST

2676 0000 CAWCHK, 0
2677 0000 CNTSHF, 0
2700 4003 PRLSW3, 4003
2701 7764 MIN14, -0014
2702 0000 PSELF, 0
2703 4000 PRCAW, 4000
2704 0000 RITBUK, 0
2705 3777 RSTMB0, 3777
2706 0756 ARING, RING

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PACFLC 0076
 ACASFT 1665
 ADHAK 0633
 ADHAKA 0624
 ADD 0243
 ADD1 1444
 ADFC 0563
 AEXIT 0344
 AGAIN 1471
 AINTON 0100
 ALFTTR 0552
 ALLCLK 0352
 ALSRA 1366
 ALSRF 1166
 ALSRJ 1365
 ALSRTS 0755
 ANOP 2260
 ANSR 0623
 ARING 2706
 ARSTR 0077

 ASCI 0562
 ASCII 0630
 ASURTR 0626
 ATABLE 0441
 ATS681 1515
 ATTITS 1764
 ATYPF 0632
 AT681G 2135
 AT681I 2370
 AT681K 2532
 AUXAC 0141
 AWAY 0627
 BADINT 0160
 BELCHA 0773
 BFILL 0014
 BITRUK 2704
 CANCHK 2676
 CANERR 2123
 CHKCAW 2453
 CHKINT 0153
 CHKPAS 2327
 CHTPTR 0142
 CLOCK 0137
 CLOCK1 0261
 CLOCK2 0275
 CLOCK3 0311
 CLOCK4 0325
 CLOCKK 0341
 CLRFIG 0232
 CLSURR 0005
 CNTM1 2371
 CNTSHE 2677
 COMCLK 0400
 CONVRT 0641
 COUNT 0006
 CR 0522
 CRIF 0514
 CWJMP1 2033
 CW4000 2256

CW52 1363
 C125 1325
 C177 0754
 C177D 1123
 C177K 1513
 C400 0751
 C401 1725
 C5252 1762
 C600 0752
 C7777 1021
 CCONTNU 1075
 DEC 0600
 DISARL 0460
 DNEXT 1106
 DSIR 0724
 ECT 0512
 ECT1 0206
 ECT2 0072
 ECT3 0134
 ECT4 0257
 ECT5 0466
 ELSR10 1431
 ELSR11 1500
 ENDTST 2674
 ENEXT1 1140
 ENEXT2 1151

 ERACD 1111
 ERACF 1154
 ERACF 1225
 ERACG 1260
 ERACH 1313
 ERACI 1351
 ERAC1 1113
 ERAC2 1156
 ERAC3 1227
 ERAC4 1262
 ERAC5 1315
 ERAC6 1353
 ERCL1 0067
 ERCL2 0101
 ERCL3 0110
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